
Data Hiway, Box/Slot, and Data Point Form Instructions

HG12-500

Implementation
Hiway Gateway - 1

Data Hiway, Box/Slot, and Data Point Form Instructions

HG12-500
9/95

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About This Publication

This publication contains instructions for filling out the *Hiway Gateway Library Forms* in the *Implementation/Configuration Forms* binder. Specific instructions are provided on filling out the Hiway and box/slot forms (SW88-530 through SW88-539). General instructions are provided for filling out the process data point forms (HG88-501 through HG88-532) and specific instructions for filling out these forms are provided in the *Hiway Gateway Parameter Reference Dictionary* in the *Implementation/Hiway Gateway - 1* binder.

Before you use this publication, you should be familiar with the information in the *Configuration Data Collection Guide* in the *Implementation/Startup & Reconfiguration - 2* binder.

This publication supports TDC 3000^X software release 500.

Change bars are used to indicate paragraphs, tables, or illustrations containing changes that have been made by Document Change Notices or an update. Pages revised only to correct minor typographical errors contain no change bars.

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INTRODUCTION

Section 1

This manual provides instructions on how to fill out the *HG88-500* Data Hiway, Box/Slot, and Data Point forms. These forms are located in the *Implementation/Configuration* binder and correspond to displays that are called up under the Hiway Gateway selection on the TDC 3000 Engineering Personality Menu.

1.1 GENERAL FORM INSTRUCTIONS

1.1.1 Line Numbers

The Hiway Gateway forms have been designed so that the lines on them correspond with those on the Parameter Entry Displays (PEDs) at the Universal Station. The Parameter Entry Displays will subsequently be used to enter the information from the form into the system. The lines on each form are consecutively numbered. In some cases, the line numbers may contain an alphabetic suffix letter. An alphabetic suffix indicates that the line may be exposed or suppressed on the PED, depending on the selection made in a previous line. This is explained further in sections that discuss each type of form.

1.1.2 Defaults

Default values are values that are assigned by the system. If you choose not to make an entry on a particular line, the default value is used. The default value for each parameter is listed on the configuration form, in the form instructions, and on the data-entry displays.

1.1.3 Pick and Port Notation

The forms have two types of entries:

1. Picks (also called touch-targets)—For selecting options. All options are on the forms. For each such entry, circle one of the options.
2. Ports—For entering data. The form shows the maximum number of characters that you can enter and the set of valid characters; write in an alphanumeric character-string.

1.2 REFERENCES

The following publications contain useful information that you may need when filling out the forms—

<u>Title</u>	<u>Publication No.</u>	<u>Binder</u>
Basic Controller, Algorithms Engineering Data	CB-09-01	Basic Controller Product Manual/25-001
Configuration Data Collection Guide	SW12-500	Implementation/Startup & Reconfiguration - 2
Data Entity Builder manual	SW11-511	Implementation/Engineering Operations - 1
Engineer's Reference Manual *	SW09-505	Implementation/Startup & Reconfiguration - 2
Extended Controller, Algorithms Engineering Data	CB-10-09	Extended Controller Product Manual/25-023
Hiway, Box Slot, and Data Point Forms	HG88-500	Implementation/Configuration Forms
Hiway Gateway Control Functions	HG09-501	Implementation/Hiway Gateway - 1
Hiway Gateway Parameter Reference Dictionary	HG09-540	Implementation/Hiway Gateway - 1
Multifunction Controller, Algorithms Engineering Data	BC-10-01	Multifunction Controller Product Manual/25-027
Network Forms Instructions	SW12-505	Implementation/Startup & Reconfiguration - 1
System Control Functions	SW09-501	Implementation/Startup & Reconfiguration - 2

* Section 8 of the Engineer's Reference Manual provides configuration guidelines that supplement the Hiway and Box/Slot form instructions that appear in Part 2 of this manual.

1.3 CONFIGURATION LIMITS/RESTRICTIONS

The configuration limits/restrictions for each Hiway Gateway in the system are listed in Table 1-1.

Table 1-1 — Hiway Gateway Configuration Limits/Restrictions

LIMIT/RESTRICTION PER HG	AFFECTED PARAMETER
Point Assignment: 3000 total data points 3000 analog inputs (includes analog composites and regulatories) 500 CB RVs 100 HLPIU control counters Input and output portion of a composite data point must reside in the same box.	BOXNUM, OUTBOXNM
PV Configuration: Calibration offset can be used for 250 analog inputs (including analog composites, regulatories, and CB RVs)	CALIBOFF
Alarming: 50 data points with emergency alarm priority 100 CB RV data points can be configured for alarming 500 contact-cutout secondary data points 600 data points can have event-initiated processing	ALPRIOR ALFMT CCRANK EIPPCODE
Operating Configuration: 500 digital data points whose output action is momentary (doorbell)	OUTIND, MOOUTIND

HIWAY AND BOX/SLOT FORM INSTRUCTIONS Section 2

2.1 DATA HIWAY AND BOX SLOT CONFIGURATION

In general, Data Hiway and Box/Slot configuration provides two functions:

1. It establishes a basis for communication between LCN Nodes (such as a Universal Station) and Data Hiway boxes (such as BASIC Controllers) through Hiway Gateways, which are both LCN Nodes and Data Hiway boxes. This basis for communications depends on
 - assignment of Hiway Numbers to HG nodes on an LCN,
 - assignment of Data Hiway addresses to each box (including HGs) on each physical Data Hiway, and
 - assignment of each box on each physical Data Hiway to each HG on that physical Data Hiway, according to the function of the HG in relation to the box.

These assignments and the relationships between them are described under the following three headings and are illustrated in Figure 2-1.

2. It establishes part of the data base that is required to define the functioning of each box on each Data Hiway. Related information that provides details for configuring box data bases is given in the Hiway Gateway Configuration publications.

2.1.1 Data Hiway Number Assignment

Data Hiway numbers are assigned to Data Hiways by associating them with pairs of Hiway Gateways during LCN Node configuration. Up-to-20 Data Hiways can be configured to each LCN by assigning Hiway Numbers that are unique on that LCN.

Data Hiway number assignment is a part of Network configuration. Because modification of the Network Configuration file requires system shutdown/startup of all LCN nodes, Hiway Number assignment should be carefully planned.

A single physical Data Hiway, which has all the physical characteristics and requirements normally associated with Data Hiways, can be treated as two or more functionally separate Data Hiways. As shown in Figure 2-1, a single physical Data Hiway can interface to an LCN through two or more HGs, each of which has a Hiway Number unique on that LCN. Also, a single physical Data Hiway can interface to two or more LCNs. In this case, the Hiway Numbers assigned to the other LCN must be unique on that LCN, but need not be different from Hiway Numbers assigned to different LCNs. (For example, Hiway NN on LCN B in Figure 2-1 may also be assigned Hiway Number 01, which is assigned to LCN A.) The functionally separate Data Hiways are shown in Figure 2-1 by dashed lines that indicate the boundaries of each functional Data Hiway. The method of assigning boxes to each functional Data Hiway is described under heading 2.1.3.

2.1.2 Data Hiway Address Assignment (Box Number Assignment)

Each box (including HGs) on a physical Data Hiway is assigned a unique address (Box Number). An HG pair requires two consecutive Box Numbers, which are always even/odd for primary/redundant HGs. In addition, certain types of process boxes require special assignment within the box number range. The following is a summary of these restricted address assignments:

HLPIU, LLPIU, LEPIU, DHP	5 to 31 (2nd address = 1st plus 32)
MC or A-MC	5 to 31 (2nd address = 1st plus 32)
EC, CB	5 to 63
HG (recommended)	2/3 (first pair), 32/33 (2nd) 34/35 (3rd)
H4500 with HIM	1
OP STNs, GPCIs, or PCSIs	4 (all)
620 HIM	5 to 31 (2nd address = 1st plus 32).

A single 620 HIM can have 2, 4, 6, or 8 box numbers, depending on the number of slots to be configured, as follows:

<i>Box Numbers Required (Total)</i>	<i>Number of Slots</i>
2	30
4	60
6	90
8	120

Each set of 2 box numbers (initial box number N, and N+32) can contain 30 slots, as shown in the above listing. The initial box number N (1st address), has a range of 5 to 31. When using more than 2 box numbers for a 620 HIM, the initial (1st address) box numbers must be consecutive starting with N, and the higher-order box numbers (2nd address) must be consecutive starting with N+32. As an example, assume that there are 8 box numbers to be configured and that the four lower-order box numbers starting with N are: N, N+1, N+2, and N+3. The four consecutive higher-order box numbers are then automatically assigned when the user enters Extended for the DHP Box Size (BOXSIZE) parameter.

No duplicate addresses are allowed on the Data Hiway, except for Operator Stations, General Purpose Computer Interfaces (GPCIs) and Personal Computer Serial Interfaces (PCSIs).

2.1.3 HG/Box Assignments

Each box (including HGs) on a physical Data Hiway must be configured to let other boxes know of its existence and also to let them know of its assigned function. The HG/Box assignments are relative; that is, they define the operational capabilities of each box relative to each functional Data Hiway. There are four types of HG/Box assignments:

1. ThisHG This assignment means that an HG allows normal AM communications and other typical HG/process box functions for a box, such as:
 - building a process data base in the box
 - Time-Out Gate processing (AM control)
 - data-access operations (reads/writes between boxes and the LCN)
 - checkpointing and loading the box data base
 - change detection (and setting) - (configurable)
 - event processing (communication of state changes and alarms) - (configurable)
 - complete Hiway Security operation

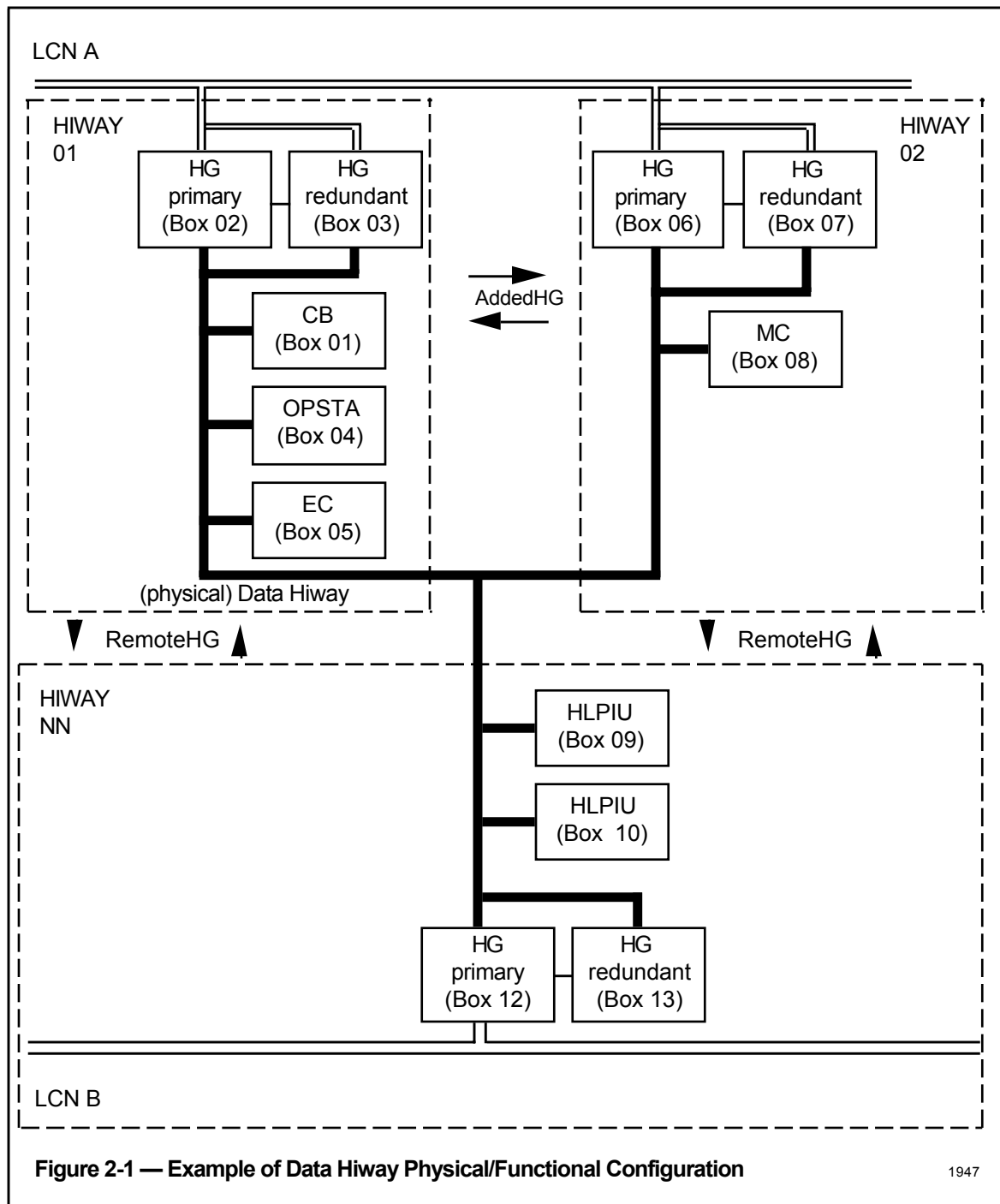
2. AddedHG This assignment blocks communications between nodes on an LCN and the box being configured, allowing only
 - Hiway Security checking through the addressing diagnostics
3. RemoteHG This assignment enables limited communications between nodes on an LCN and the box being configured, allowing
 - the same as ThisHG except no AM control
4. H4500 This assignment enables limited communications between nodes on an LCN and the box being configured, allowing
 - the same as ThisHG except no AM control

When a physical Data Hiway includes more than one HG pair (and therefore, more than one Hiway Number Assignment), each box on the physical Data Hiway must be separately configured, once for each HG pair on that physical Data Hiway. For example, all thirteen boxes shown in Figure 2-1 must be separately configured for Hiway 01, for Hiway 02, and for Hiway NN.

The HG/Box assignments are relative in that, depending on its function, a given box can at the same time be assigned as ThisHG, and AddedHG, and RemoteHG. For example, Box 02 in Figure 2-1 is ThisHG on Hiway 01, but is an AddedHG to Hiway 02. Similarly, Box 06 is ThisHG on Hiway 02, but is AddedHG relative to Hiway 01. The Box-assignment designations between the dashed lines that show Hiway boundaries in Figure 2-1 and the accompanying arrows indicate the relationship of each box in each hiway to all other boxes in all other hiways on the same Physical Data Hiway. Figures 2-2 and 2-3 include lists that detail the configuration requirements necessary to define each box shown in Figure 2-1 in terms of Hiway Number, Box Number, Box Type, and Box Assignment.

NOTE

If Uninterrupted Automatic Control (UAC) is included, it must be configured entirely on the same functional Data Hiway; that is, the Reserve Controller Director and all controllers (Basic Controllers, Extended Controllers, or Multifunction Controllers, both primary and reserve) must be assigned on ThisHG to the same Hiway Gateway.



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A. LCN A

Form R500 SW88-530

Hiway Number 01
 HTD Assignment ThisHG
 Hiway Number 02
 HTD Assignment AddedHG

Forms SW88-531 through SW88-530

Hiway Number	01	01	01	01	01
Box Number	01	02	03	04	05
Box Type	CB	HG	HG	OPSTA	EC
Box Assignment	ThisHG	ThisHG	ThisHG		ThisHG

Hiway Number	01	01	01
Box Number	06	07	08
Box Type	HG	HG	MC
Box Assignment	AddedHG	AddedHG	AddedHG

Hiway Number	01	01	01	01
Box Number	09	10	12	13
Box Type	HLPIU	LLPIU	HG	HG
Box Assignment	RemoteHG	RemoteHG	RemoteHG	RemoteHG

Hiway Number	02	02	02	02	02
Box Number	01	02	03	04	05
Box Type	CB	HG	HG	OPSTA	EC
Box Assignment	AddedHG	AddedHG	AddedHG		AddedHG

Hiway Number	02	02	02
Box Number	06	07	08
Box Type	HG	HG	MC
Box Assignment	ThisHG	ThisHG	ThisHG

Hiway Number	02	02	02	02
Box Number	09	10	12	13
Box Type	HLPIU	LLPIU	HG	HG
Box Assignment	RemoteHG	RemoteHG	RemoteHG	RemoteHG

Figure 2-2 — Example of Data Hiway Physical/Functional Configuration

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B. LCN B

Form R500 SW88-530

Hiway Number nn
 HTD Assignment ThisHG

Forms SW88-531 through SW88-530

Hiway Number	nn	nn	nn	nn	nn
Box Number	01	02	03	04	05
Box Type	CB	HG	HG	OPSTA	EC
Box Assignment	RemoteHG	RemoteHG	RemoteHG		RemoteHG

Hiway Number	nn	nn	nn
Box Number	06	07	08
Box Type	HG	HG	MC
Box Assignment	RemoteHG	RemoteHG	RemoteHG

Hiway Number	nn	nn	nn	nn
Box Number	09	10	11	12
Box Type	HLPIU	LLPIU	HG	HG
Box Assignment	ThisHG	ThisHG	ThisHG	ThisHG

Figure 2-3 — Example of Data Hiway Physical/Functional Configuration

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2.1.4 Filling In Hiway/Box Slot Forms

The instructions for filling in the forms are in tabular format. Each line on the form has a number that corresponds to a number in the table for that form, where the choices and instructions are given.

Some of the entries require additional entries. The relationship between the line(s) that require additional entries and the line(s) on which the additional entries are made is indicated in one of two ways:

- by alphabetic suffixes, when the selectively disclosed parameters appear on the displays immediately after the parameter that requires more entries. For example, Lines 8A, 8B, and 8C depend on the entries made in line 8.
- by asterisks, two (**) for the parameters that require additional entries and three (***) for the selectively disclosed parameters. For example, Lines 42A-42F (***) depend on the entries made in lines 8 (**) and 9 (**).

The forms have two other types of entries:

- Options—Sometimes you must select an option. All options are on the forms. For each such entry, circle one of the options.
- Data—sometimes the form provides a series of blank spaces where can enter numbers or a characters. The form or form instructions tell what characters are valid and the maximum number characters that can be entered. Write in an alphanumeric character-string.

HIWAY CONFIGURATION FORM

HG-HIWAY POINT ASSIGNMENT DISPLAY

PAGE 1

1 HIWAY NUMBER (Default = 01)
(HWYNUM)

2 HTD ASSIGNMENT THISHG REMOTEHG HTD ADDEDHG
(HWYHTD) (Default = HTD)

2A SWITCH INTERVAL (MIN) (Default = 1440)
(HWYSWINT)

2B SWITCH BASE TIME (MIN) (Default = 0)
(HWYSWBAS)

3 SOE SYNCHRONIZATION DISABLE ENABLE (Default = DISABLE)
(SOESYNCH)

4 COMMAND DISAGREE TEXT (Default = CMD_DIS)
(CDTEXT)

DUAL DIGITAL INPUT NONCOMPLEMENTARY TEXT

5 TEXT1 (Default = BADPV)
(NCMPTXT1)

6 TEXT2 (Default = INBETWN)
(NCMPTXT2)

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.1 Data Hiway Configuration Form SW88-530

Table 2-1 contains instructions for Configuration Form *SW88-530*.

Table 2-1 — Instructions for Data Hiway HG-Hiway Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG-HIWAY POINT ASSIGNMENT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN-node configuration.*
2	HTD ASSIGNMENT (HWYHTD)	Circle one: THISHG REMOTEHG HTD ADDEDHG (Default = HTD)	Defines which device performs the functions of a Hiway Traffic Director. ThisHG = the HG assigned to this Hiway Number* acts as HTD on the physical Data Hiway. RemoteHG = The HG on another LCN acts as HTD. Added HG = HG on this LCN, but assigned to a different Hiway Number acts as HTD If THISHG is entered, also make entries in Lines 2A and 2B, then go to Line 3. If REMOTEHG or ADDEDHG is entered, go to Line 3.
2A	SWITCH INTERVAL (MIN) (HWYSWINT)	Enter a number from 1 to 1440 (Default = 1440)	Defines the amount of time (in minutes) between automatic switching of the primary and backup Data Hiways; 1440 minutes = number of minutes in one day.
2B	SWITCH BASE TIME (MIN) (HWYSWBAS)	Enter the time of day (in minutes) that automatic hiway switching is to take place (Default = 0)	Example: 1:30 A.M. would be entered as 90. Example: 3:00 P.M. would be entered as 900.

(Continued)

*Refer to Network Form SW88-509, Hiway Gateway Node Display

HIWAY CONFIGURATION FORM

HG-HIWAY POINT ASSIGNMENT DISPLAY

PAGE 1

1 HIWAY NUMBER (Default = 01)
(HWYNUM)

2 HTD ASSIGNMENT THISHG REMOTEHG HTD ADDEDHG
(HWYHTD) (Default = HTD)

2A SWITCH INTERVAL (MIN) (Default = 1440)
(HWYSWINT)

2B SWITCH BASE TIME (MIN) (Default = 0)
(HWYSWBAS)

3 SOE SYNCHRONIZATION DISABLE ENABLE (Default = DISABLE)
(SOESYNCH)

4 COMMAND DISAGREE TEXT (Default = CMD_DIS)
(CDTEXT)

DUAL DIGITAL INPUT NONCOMPLEMENTARY TEXT

5 TEXT1 (Default = BADPV)
(NCMPTXT1)

6 TEXT2 (Default = INBETWN)
(NCMPTXT2)

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-1 — Instructions for Data Hiway HG-Hiway Configuration Form (Continued)

Table 2-1 Instructions for Data Hiway/HG Hiway Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS											
ASSIGNMENT DISPLAY														
3	SOE SYNCHRONIZATION (SOESYNCH)	Circle one: DISABLE ENABLE (Default = DISABLE)	Select ENABLE if the HG being configured is to generate SOE (Sequence of Events) words for HLPIUs on this hiway. If an H4500 is on this Hiway, it must generate SOE words											
4	COMMAND DISAGREE TEXT (CDTEXT)	If this Data Hiway includes Multifunction Controllers, enter up-to-8 characters A–Z, 0–9, and space ! " % & ' () * + - / : ; > < = ? , . \$ (Default = CMD DIS)	This descriptor is displayed if an MC has been configured for Command Disagree and if both inputs of a dual digital data point are in the same state.											
5	TEXT 1 (NCMPTXT1)	Enter a descriptor of up-to-8 characters A–Z, 0–9, and space ! " % & ' () * + - / : ; > < = ? , . \$ (Default = BADPV)	This descriptor is displayed when TEXT 1 is selected for ZZTEXT during configuration of a point in a box assigned to the same HG and both inputs to a dual digital point are zero (00). If both inputs are ones (11), the TEXT 2 descriptor is displayed.											
6	TEXT 2 (NCMPTXT2)	Enter a descriptor of up-to-8 characters A–Z, 0–9, and space ! " % & ' () * + - / : ; > < = ? , . \$ (Default = INBETWN)	This descriptor is displayed when TEXT 2 is selected for ZZTEXT during configuration of a point in a box assigned to the same HG and both inputs to a dual digital point are zero (00). If both inputs are ones (11), the TEXT 1 descriptor is displayed. Selection of Text Descriptors (ZZTEXT selection, part of point configuration) <table><tr><td rowspan="2">ZZTEXT Selection</td><td colspan="2">Non-Complementary Input Sales</td></tr><tr><td>00</td><td>11</td></tr><tr><td>TEXT 1</td><td>TEXT 1</td><td>TEXT 2</td></tr><tr><td>TEXT 2</td><td>TEXT 2</td><td>TEXT 1</td></tr></table> Descriptor displayed	ZZTEXT Selection	Non-Complementary Input Sales		00	11	TEXT 1	TEXT 1	TEXT 2	TEXT 2	TEXT 2	TEXT 1
ZZTEXT Selection	Non-Complementary Input Sales													
	00	11												
TEXT 1	TEXT 1	TEXT 2												
TEXT 2	TEXT 2	TEXT 1												

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HG BOX DATA POINT CONFIGURATION FORM

HG BOX POINT DISPLA

1	HIWAY NUMBER (HWYNUM)		(Default = 1
2	BOX NUMBER (BOXNUM)		(Default = 0
3	BOX TYPE (BOXTYPE)	CB AU HLPIU L LEPIU CBRCD EC E MC MCRCD DHP O H4500 HG GPCI NOTCONFG (Ente	
4	BOX ASSIGNMENT (BOXASSN)	THISHG REMOTEHG ADDEDHG	(Default = TH
5	EVENT PROCESSING (EVENTPRC)	DISABLE	(DEFAULT = DISABLE ONLY FOR THIS BOX

1	HIWAY NUMBER (HWYNUM)		(Default = 1
2	BOX NUMBER (BOXNUM)		(Default = 0
3	BOX TYPE (BOXTYPE)	CB AU HLPIU L LEPIU CBRCD EC E MC MCRCD DHP O H4500 HG GPCI NOTCONFG (Ente	
4	BOX ASSIGNMENT (BOXASSN)	THISHG REMOTEHG ADDEDHG	(Default = TH
5	EVENT PROCESSING (EVENTPRC)	DISABLE	(DEFAULT = DISABLE ONLY FOR THIS BOX

1	HIWAY NUMBER (HWYNUM)		(Default = 1
2	BOX NUMBER (BOXNUM)		(Default = 0
3	BOX TYPE (BOXTYPE)	CB AU HLPIU L LEPIU CBRCD EC E MC MCRCD DHP O H4500 HG GPCI NOTCONFG (Ente	
4	BOX ASSIGNMENT (BOXASSN)	THISHG REMOTEHG ADDEDHG	(Default = TH
5	EVENT PROCESSING (EVENTPRC)	DISABLE	(DEFAULT = DISABLE ONLY FOR THIS BOX

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.2 HG Box Data Point (Box/Slot) Configuration Form SW88-531

Table 2-2 contains instructions for Configuration Form *SW88-531*.

Table 2-2 — Instructions for HG Box Data Point Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG BOX POINT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN node configuration*.
2	BOX NUMBER (BOXNUM)	Enter a box number from 1 to 63. Box number 4 is generally reserved for the Operator Station if an Operator Station is on this hiway (Default = 0)	Defines the box address of the HG on the selected Data Hiway. If there is no operator station on this hiway, box number 4 can be assigned to the HG.
3	BOX TYPE (BOXTYPE)	Enter HG	Defines this box as an HG.
4	BOX ASSIGNMENT (BOXASSN)	Circle one: THISHG REMOTEHG ADDEDHG (Default = THISHG)	Defines the function of the HG being configured relative to this Data Hiway (see heading 2.1.3). –ThisHG = The HG being configured is assigned to the Hiway Number entered on Line 1. –RemoteHG = The HG being configured is assigned to a Hiway Number on a different LCN. –Added HG = The HG being configured is assigned to a Hiway Number different from that entered on line 1.
5	EVENT PROCESSING (EVENTPRC)	Default = Disable	Defines whether process events that pertain to this parameter hiway box are to be routed through the HG to the LCN Disable = Process events are not routed through the HG to the LCN.

*Refer to Network Form SW88-509, Hiway Gateway Node Display

CB BOX DATA POINT CONFIGURATION FORM

HG CB BOX POINT DISPL

1	HIWAY NUMBER		(Default = 1
	(HWYNUM		
2	BOX NUMBER		(Default = 0
	(BOXNUM		
3	BOX TYPE	CB AU HLPIU L	
	(BOXTYPE)	LEPIU CBRCD EC E	
		MC MCRCD DHP O	
		H4500 HG GPCI NOTCONFIG (Ente	
4	BOX ASSIGNMENT	THISHG REMOTEHG ADDEDHG H4500 (Defa	
**	(BOXASSN)		TH
5	LOAD DESTINATION	HG HG HIWAY (Default =	
**	(LOADDEST		
6	CHANGE DETECTION	NOTCONFIG DETECT SET (Default = NOTC	
	(CHNGFLAG		
7	EVENT PROCESSING	DISABLE ENABLE (Default = DIS	
	(EVENTPRC		
8	BOX TRENDING	NOTREND TREND (Default = NOTR	
	(BOXTREND		
9	TOG INTERVAL 1		(Default = 0 seconds
***	(BOXTOG1		
10	TOG INTERVAL 2		(Default = 0 seconds
***	(BOXTOG2		
10A	SLOT 1 TOG INTERVAL	INTERV1 INTERV2 (Default = INT	
	(TOGINTSL		
10B	SLOT 2 TOG INTERVAL	INTERV1 INTERV2 (Default = INTE	
19C	SLOT 3 TOG INTERVAL	INTERV1 INTERV2 (Default = INT	
10D	SLOT 4 TOG INTERVAL	INTERV1 INTERV2 (Default = INTE	
10E	SLOT 5 TOG INTERVAL	INTERV1 INTERV2 (Default = INT	
10F	SLOT 6 TOG INTERVAL	INTERV1 INTERV2 (Default = INTE	
10G	SLOT 7 TOG INTERVAL	INTERV1 INTERV2 (Default = INT	
10H	SLOT 8 TOG INTERVAL	INTERV1 INTERV2 (Default = INTE	

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.3 CB Box Data Point (Box/Slot) Configuration Form SW88-532

Table 2-3 contains instructions for Configuration Form *SW88-532*.

Table 2-3 — Instructions for CB Box Data Point Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG CB BOX POINT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN node configuration.
2	BOX NUMBER (BOXNUM)	Enter a number from 1 to 63 (Default = 0)	Defines the hiway address of this CB box on the Data Hiway.
3	BOX TYPE (BOXTYPE)	Enter CB	
4 **	BOX ASSIGNMENT (BOXASSN)	Circle one: THISHG REMOTEHG ADDEDHG H4500 (Default = THISHG)	Defines the function relative to this CB of <u>each</u> HG on the same physical Data Hiway. Refer to Table 2-2 and to heading 2.1.3. If AddedHG is entered, configuration is complete.
5 **	LOAD DESTINATION (LOADDEST)	Circle one: HG HG_HIWAY (Default = HG)	If ThisHG was entered on line 4 and HG_HIWAY is entered on line 5, fill in the rest of this form, including lines 9 through 10H, as applicable. If HG is entered on line 5, fill in only lines 6 through 8.
6	CHANGE DETECTION (CHNGFLAG)	Circle one: NOTCONFIG DETECT SET (Default = NOTCONFIG)	Select: –NOTCONFIG if no preferred devices are on this Data Hiway (Speeds up communication.) –DETECT to enable HG to alert LCN devices such as other HGs or H4500s to changes in CB status that originated in preferred devices on the same Data Hiway.

(Continued)

*Refer to Network Form SW88-509, Hiway Gateway Node Display

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CB BOX DATA POINT CONFIGURATION FORM

HG CB BOX POINT DISPL

1	HIWAY NUMBER		(Default = 1
	(HWYNUM		
2	BOX NUMBER		(Default = 0
	(BOXNUM		
3	BOX TYPE	CB AU HLPIU L	
	(BOXTYPE)	LEPIU CBRCD EC E	
		MC MCRCD DHP O	
		H4500 HG GPCI NOTCONFIG (Ente	
4	BOX ASSIGNMENT	THISHG REMOTEHG ADDEDHG H4500 (Defa	
**	(BOXASSN)		TH
5	LOAD DESTINATION	HG HG HIWAY (Default =	
**	(LOADDEST		
6	CHANGE DETECTION	NOTCONFIG DETECT SET (Default = NOTC	
	(CHNGFLAG		
7	EVENT PROCESSING	DISABLE ENABLE (Default = DIS	
	(EVENTPRC		
8	BOX TRENDING	NOTTREND TREND (Default = NOTR	
	(BOXTREND		
9	TOG INTERVAL 1		(Default = 0 seconds
***	(BOXTOG1		
10	TOG INTERVAL 2		(Default = 0 seconds
***	(BOXTOG2		
10A	SLOT 1 TOG INTERVAL	INTERV1 INTERV2 (Default = INT	
	(TOGINTSL		
10B	SLOT 2 TOG INTERVAL	INTERV1 INTERV2 (Default = INTE	
19C	SLOT 3 TOG INTERVAL	INTERV1 INTERV2 (Default = INT	
10D	SLOT 4 TOG INTERVAL	INTERV1 INTERV2 (Default = INTE	
10E	SLOT 5 TOG INTERVAL	INTERV1 INTERV2 (Default = INT	
10F	SLOT 6 TOG INTERVAL	INTERV1 INTERV2 (Default = INTE	
10G	SLOT 7 TOG INTERVAL	INTERV1 INTERV2 (Default = INT	
10H	SLOT 8 TOG INTERVAL	INTERV1 INTERV2 (Default = INTE	

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-3 — Instructions for CB Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG CB BOX POINT DISPLAY			
7	EVENT PROCESSING (EVENTPRC)	Circle one: DISABLE ENABLE (Default = DISABLE)	Select ENABLE if process alarms, process changes, and operator messages are to be routed through this HG to the LCN.
8	BOX TRENDING (BOXTREND)	Circle one: NOTREND TREND (Default = NOTREND)	Defines whether this box has the trend option.
9 ***	TOG INTERVAL 1 (BOXTOG1)	Enter 0 to 9999 seconds	TOG (Time-Out Gate) intervals are user-specified, selectable maximum times between successive transmissions from a supervisory controller to a particular slot in a box. If the selected interval is exceeded, the slot sheds from its remote cascade mode. If both entries are zero, no selection is provided; skip lines 10A through 10H.
10 ***	TOG INTERVAL 2 (BOXTOG2)	0 = Never shed (Default = 0 seconds)	
10A – 10H	SLOT 1–8 TOG INTERVAL (TOGINTSL)	Circle one: INTERV1 INTERV2 (Default = INTERV1)	Specifies whether TOG INTERVAL 1 (INTERV1) or TOG INTERVAL 2 (INTERV2) entered on Lines 9 and 10 is to be the time-out gate interval for the particular slot.

DHP/620-HIM BOX DATA POINT CONFIGURATION FORM

HG DHP BOX POINT DISPLA

1	HIWAY NUMBER		(Default = 1
	(HWYNUM		
2	BOX NUMBER		(Default = 0
	(BOXNUM		
3	BOX TYPE	CB AU HLPIU L	
	(BOXTYPE)	LEPIU CBRCD EC E	
		MC MCRCD DHP O	
		H4500 HG GPCI NOTCONFG (Enter	
		(Default = THIS	
4	BOX ASSIGNMENT	THISHG REMOTEHG ADDEDHG	(Default = TH
**	(BOXASSN)	H45	
5	LOAD DESTINATION	HG HG HIWAY	(Default =
**	(LOADDEST		
6	DHP BOX SIZE	REGULAR EXTENDED	(Default = REGU
	(BOXSIZE		
7	BOX PROTOCOL	ALLENBRD MODICON HONYWELL	(Default = ALLE
	(BOXPROT		
8	BOX STARTUP	COLD HOT	(Default = C
	(BOXSTART		
9	SCAN TIME (SEC)		(Default = 0
	(SCANTIME		
10	PC1 MODEL TYPE	NOTCONFG M384 M484 M584	(For Mod
	(PC1TYPE)	(Default = NOTCONFG)-----	
		NOTCONFG APLC APLC2 APL	
		APLC220 APLC230 AMINPLC2	(For Allen Bra
		(Default = NOTCONFG)-----	
		NOTCONFG IPC620 (For Honeyw	
		(Default = NOTCONFG)-----	
10A	PORT NUMBER		(Default = 1
	(PC1PORT		
10B	PORT ADDRESS		(Default = 0
	(PC1PORTA		
10C	KEEP ALIVE ADDRESS		(Default = 0
	(PC1ALIVE		
10D	KEEP ALIVE BIT POSITIO		(Default = 0
	(PC1ALVBT		
10E	SPECIFIER		(Default = 0
	(PC1ALVSP		
11	PC2 MODEL TYPE	NOTCONFG M384 M484 M584	(For Mod
	(PC2TYPE)	-----	
		NOTCONFG APLC APLC2 AP	
		APLC220 APLC230 AMINPLC2	(For Allen Bra

		NOTCONFG IPC620 (For Honeyw	

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.4 DHP/620-HIM Box Data Point (Box/Slot) Configuration Form SW88-533

Table 2-4 contains instructions for Configuration Form *SW88-533*.

Table 2-4 — Instructions for DHP/620-HIM Box Data Point Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG DHP BOX POINT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN node configuration.*
2	BOX NUMBER (BOXNUM)	Enter a number from 1 to 53 (box number 4 is reserved for the Operator Station) (Default = 0)	Defines the box address of this DHP (or 620 HIM) box on the selected Data Hiway. A single 620 HIM contains 30-120 slots depending on whether 2, 4, 6, or 8 box numbers are used. The user must fill out a separate SW88-533 configuration form for each of the four lower-order (1st address) numbers. Refer to IPCD Publication 620-8981, 620-0081 Hiway Interface Module User Manual for the location of the 620 HIM Base Address (1st address) and Block Enable (number of boxes) settings.
3	BOX TYPE (BOXTYPE)	Enter DHP	Defines this box as being the Data Hiway Port (DHP).
4 **	BOX ASSIGNMENT (BOXASSN)	Circle one: THISHG REMOTEHG ADDEDHG H4500 (Default = THISHG)	Defines the function relative to this DHP of each HG on the same physical Data Hiway. (Refer to heading 2.1.3.) If Added HG is entered, configuration is complete.
5 **	LOAD DESTINATION (LOADDEST)	Circle one: HG HG_HIWAY (Default = HG)	If This HG was entered on line 4, and HG_HIWAY is entered on line 5, fill in the rest of this form, including lines 50–51AO, as applicable. Otherwise, fill in the rest of this form except for lines 50–51AO.

(Continued)

*Refer to Network Form SW88-509, Hiway Gateway Node Display

DHP/620-HIM BOX DATA POINT CONFIGURATION FORM

HG DHP BOX POINT DISPLA

1	HIWAY NUMBER		(Default = 1
	(HWYNUM		
2	BOX NUMBER		(Default = 0
	(BOXNUM		
3	BOX TYPE	CB AU HLPIU L	
	(BOXTYPE)	LEPIU CBRCD EC E	
		MC MCRCD DHP O	
		H4500 HG GPCI NOTCONFG (Enter	
		(Default = THIS	
4	BOX ASSIGNMENT	THISHG REMOTEHG ADDEDHG	(Default = TH
**	(BOXASSN)	H45	
5	LOAD DESTINATION	HG HG HIWAY	(Default =
**	(LOADDEST		
6	DHP BOX SIZE	REGULAR EXTENDED	(Default = REGU
	(BOXSIZE		
7	BOX PROTOCOL	ALLENBRD MODICON HONYWELL	(Default = ALLE
	(BOXPROT		
8	BOX STARTUP	COLD HOT	(Default = C
	(BOXSTART		
9	SCAN TIME (SEC)		(Default = 0
	(SCANTIME		
10	PC1 MODEL TYPE	NOTCONFG M384 M484 M584	(For Mod
	(PC1TYPE)	(Default = NOTCONFG)-----	
		NOTCONFG APLC APLC2 APL	
		APLC220 APLC230 AMINPLC2	(For Allen Bra
		(Default = NOTCONFG)-----	
		NOTCONFG IPC620 (For Honeyw	
		(Default = NOTCONFG)-----	
10A	PORT NUMBER		(Default = 1
	(PC1PORT		
10B	PORT ADDRESS		(Default = 0
	(PC1PORTA		
10C	KEEP ALIVE ADDRESS		(Default = 0
	(PC1ALIVE		
10D	KEEP ALIVE BIT POSITIO		(Default = 0
	(PC1ALVBT		
10E	SPECIFIER		(Default = 0
	(PC1ALVSP		
11	PC2 MODEL TYPE	NOTCONFG M384 M484 M584	(For Mod
	(PC2TYPE)	-----	
		NOTCONFG APLC APLC2 AP	
		APLC220 APLC230 AMINPLC2	(For Allen Bra

		NOTCONFG IPC620 (For Honeyw	

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-4 — Instructions for DHP/620-HIM Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG DHP BOX POINT DISPLAY			
6	DHP BOX SIZE (BOXSIZE)	Circle one: REGULAR EXTENDED (Default = REGULAR)	Defines whether DHP has 15 slots (REGULAR) or 30 slots (EXTENDED). For a 620 HIM, enter Extended. Slot 16 is reserved for DHP/PC configuration and box status. Slot 32 is reserved for future use. A single 620 HIM has a possible total of 120 slots; 30 slots for each set of box numbers (N and N+32).

(Continued)

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DHP/620-HIM BOX DATA POINT CONFIGURATION FORM
(Continued)

HG DHP BOX POINT DISPLA

11A	PORT NUMBER	☐	☐	(Default = 1			
	(PC2PORT						
11B	PORT ADDRESS	☐	☐	(Default = 0			
	(PC2PORTA						
11C	KEEP ALIVE ADDRESS	☐	☐	(Default = 0			
	(PC2ALIVE						
11D	KEEP ALIVE BIT POSITION	☐	☐	(Default = 0			
	(PC2ALVBT						
11E	SPECIFIER	☐	☐	(Default = 0			
	(PC2ALVSP						
12	PC3 MODEL TYPE	NOTCONFG	M384	M484	M584	(For Mod	
	(PC3TYPE)	-----					
		NOTCONFG	APLC	APLC2	AP		
		APLC220	APLC230	AMINPLC2	(For Allen Bra		

		NOTCONFG	IPC620	(For Honeyw			

12A	PORT NUMBER	☐	☐	(Default = 1			
	(PC3PORT						
12B	PORT ADDRESS	☐	☐	(Default = 0			
	(PC3PORT						
12C	KEEP ALIVE ADDRESS	☐	☐	(Default = 0			
	(PC3ALIVE						
12D	KEEP ALIVE BIT POSITION	☐	☐	(Default = 0			
	(PC3ALVBT						
12E	SPECIFIER	☐	☐	(Default = 0			
	(PC3ALVSP						
13	PC4 MODEL TYPE	NOTCONFG	M384	M484	M584	(For Mod	
	(PC4TYPE)	-----					
		NOTCONFG	APLC	APLC2	APL		
		APLC220	APLC230	AMINPLC2	(For Allen Bra		

		NOTCONFG	IPC620	(For Honeyw			

13A	PORT NUMBER	☐	☐	(Default = 1			
	(PC4PORT						
13B	PORT ADDRESS	☐	☐	(Default = 0			
	(PC4PORTA						
13C	KEEP ALIVE ADDRESS	☐	☐	(Default = 0			
	(PC4ALIVE						
13D	KEEP ALIVE BIT POSITION	☐	☐	(Default = 0			
	(PC4ALVBT						
13E	SPECIFIER	☐	☐	(Default = 0			
	(PC4ALVSP						

References Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-4 — Instructions for DHP/620 HIM Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG DHP BOX POINT DISPLAY			
7	BOX PROTOCOL (BOXPROT)	Circle one: ALLENBRD MODICON HONYWELL (Default = ALLENBRD)	Defines the type of protocol (message format) that is to be used by this DHP (or 620 HIM). For the 620 HIM, enter Honeywell. The entry on this line determines which PC models are available in Lines 10–17 (PC1–PC8 MODEL TYPE). Protocol for GE PCs is the same as that for Modicon.
8	BOX STARTUP (BOXSTART)	Circle one: HOT COLD (Default = COLD)	In a cold startup, the DHP (or 620 HIM) does not enter its normal operating state after startup and remains in a reset state. In a hot startup, the DHP (or 620 HIM) enters its normal processing state after startup, and processing is enabled.
9	SCAN TIME(SEC) (SCANTIME)	Enter 0 to 15 seconds. (Default = 0)	Assigns the scan time (in seconds) during which the DHP (or 620 HIM) updates all the PC data points in the DHP (or 620 HIM) database. A scan time of 0 seconds is considered as being free running. NOTE For a 620 HIM, make entries in Lines 10C and 10D; then go to Line 18.
10– 17 (See NOTE)	PC1–PC8 MODEL TYPE (PC1TYPE) to (PC8TYPE)	If MODICON has been entered in Line 7, circle one: NOTCONFIG M384 M484 M584 (Default = NOTCONFIG)	Defines the PC model type for up to 8 PCs (PC1–PC8). M384 = Model Type 384 M484 = Model Type 484 M584 = Model Type 584

(Continued)

NOTE

In the following instructions, similar entries are grouped to avoid cumbersome repetition. The line numbers of instructions and form entries correspond to each other, but are not shown on facing pages.

DHP/620-HIM BOX DATA POINT CONFIGURATION FORM
(Continued)

HG DHP BOX POINT DISPLA

14	PC5 MODEL TYPE (PC5TYPE)	NOTCONFIG	M384	M484	M584	(For Mod

		NOTCONFIG	APLC	APLC2	AP	
		APLC220	APLC230	AMINPLC2	(For Allen Bra	

		NOTCONFIG	IPC620	(For Honeyw		

14A	PORT NUMBER (PC5PORT	☐	(Default = 1			
14B	PORT ADDRESS (PC5PORTA	☐	(Default = 0			
14C	KEEP ALIVE ADDRESS (PC5ALIVE	☐	(Default = 0			
14D	KEEP ALIVE BIT POSITION (PC5ALVBT	☐	(Default = 0			
14E	SPECIFIER (PC5ALVSP	☐	(Default = 0			
15	PC6 MODEL TYPE (PC6TYPE)	NOTCONFIG	M384	M484	M584	(For Mod

		NOTCONFIG	APLC	APLC2	APL	
		APLC220	APLC230	AMINPLC2	(For Allen Bra	

		NOTCONFIG	IPC620	(For Honeyw		

15A	PORT NUMBER (PC6PORT	☐	(Default = 1			
15B	PORT ADDRESS (PC6PORTA	☐	(Default = 0			
15C	KEEP ALIVE ADDRESS (PC6ALIVE	☐	(Default = 0			
15D	KEEP ALIVE BIT POSITION (PC6ALVBT	☐	(Default = 0			
15E	SPECIFIER (PC6ALVSP	☐	(Default = 0			
16	PC7 MODEL TYPE (PC7TYPE)	NOTCONFIG	M384	M484	M584	(For Mod

		NOTCONFIG	APLC	APLC2	AP	
		APLC220	APLC230	AMINPLC2	(For Allen Bra	

		NOTCONFIG	IPC620	(For Honeyw		

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-4 — Instructions for DHP/620 HIM Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG DHP BOX POINT DISPLAY			
			Select M584 if newer model MODICONs are to be configured. GE Model types correspond to Modicon types as follows: GE 6/60—M384 GE 6/600—M484 GE 6/6000—M584 NOTCONFIG = Not configured APLC = Model Type PLC APLC2 = Model Type PLC2 APLC215 = Model Type PLC215 APLC220 = Model Type PLC220 APLC230 = Model Type PLC230 AMINPLC2 = Model Type Mini PLC2 If ALLENBRD has been entered in Line 7, circle one: NOTCONFIG APLC APLC2 APLC215 APLC220 APLC230 AMINPLC2 (Default = NOTCONFIG) If HONYWELL has been entered in Line 7, circle one: NOTCONFIG IPC620 (Default = NOTCONFIG) The DHP supports all Honeywell PC model types; the model type does not have to be specified further. If NOTCONFIG is selected, information—including the Keep Alive Address—for the PC will not be displayed on the DHP box status display, and PCADDRxx and SPECIFxx parameters will not be read from the 620 HIM.
10A– 17A	PORT NUMBER (PC1PORT) to (PC8PORT)	Enter a number from 1 to 4 (Default = 1)	For a 620 HIM, use the default value. Defines the DHP port number for a PC. One DHP port can handle –up-to-8 Allen Bradley or Modicon PCs –one Honeywell PC

(Continued)

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DHP/620-HIM BOX DATA POINT CONFIGURATION FORM
(Continued)

HG DHP BOX POINT DISPLA

16A PORT NUMBER | (Default = 1
 (PC7PORT)

16B PORT ADDRESS | (Default = 0
 (PC7PORTA)

16C KEEP ALIVE ADDRESS | (Default = 0
 (PC7ALIVE)

16D KEEP ALIVE BIT POSITION | (Default = 0
 (PC7ALVBT)

16E SPECIFIER | (Default = 0
 (PC7ALVSP)

17 PC8 MODEL TYPE NOTCONFIG M384 M484 M584 (For Mod
 (PC8TYPE)

 NOTCONFIG APLC APLC2 AP
 APLC220 APLC230 AMINPLC2 (For Allen Bra
 NOTCONFIG IPC620 (For Honeyw

17A PORT NUMBER | (Default = 1
 (PC8PORT)

17B PORT ADDRESS | (Default = 0
 (PC8PORTA)

17C KEEP ALIVE ADDRESS | (Default = 0
 (PC8ALIVE)

17D KEEP ALIVE BIT POSITION | (Default = 0
 (PC8ALVBT)

17E SPECIFIER | (Default = 0
 (PC8ALVSP)

18 SLOT 1 TYPE NONE DIGIN DIGOUT (Default =
 (PIUCRDY) ANALOGIN ANALOGOT COU

19 SLOT 2 TYPE NONE DIGIN DIGOUT (Default =
 ANALOGIN ANALOGOT COUN

20 SLOT 3 TYPE NONE DIGIN DIGOUT (Default =
 ANALOGIN ANALOGOT COU

21 SLOT 4 TYPE NONE DIGIN DIGOUT (Default =
 ANALOGIN ANALOGOT COUN

22 SLOT 5 TYPE NONE DIGIN DIGOUT (Default =
 ANALOGIN ANALOGOT COU

23 SLOT 6 TYPE NONE DIGIN DIGOUT (Default =
 ANALOGIN ANALOGOT COUN

24 SLOT 7 TYPE NONE DIGIN DIGOUT (Default =
 ANALOGIN ANALOGOT COU

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-4 — Instructions for DHP/620-HIM Box Data Point Configuration Form (Continued)

Table 2-4 Instructions for DHP 620 HIM Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS										
HG DHP BOX POINT DISPLAY													
10B– 17B	PORT ADDRESS (PC1PORTA) to (PC8PORTA)	Enter the device address of each PC (PC1–PC8) as follows: <table><thead><tr><th>Entry</th><th>Description</th></tr></thead><tbody><tr><td>0</td><td>Undefined</td></tr><tr><td>1–247 (Decimal)</td><td>Valid addresses for Modicon</td></tr><tr><td>1–247</td><td>Valid addresses for Honeywell</td></tr><tr><td>1–377 (Octal)</td><td>Valid addresses for Allen Bradley</td></tr></tbody></table> (Default = 0)	Entry	Description	0	Undefined	1–247 (Decimal)	Valid addresses for Modicon	1–247	Valid addresses for Honeywell	1–377 (Octal)	Valid addresses for Allen Bradley	For a 620 HIM use the default value. Defines the address of a PC on a DHP port. A zero (0) entry indicates that the PC is not configured. Range is from 0 to 255, but only entries in the listed ranges are accepted during a configuration load.
Entry	Description												
0	Undefined												
1–247 (Decimal)	Valid addresses for Modicon												
1–247	Valid addresses for Honeywell												
1–377 (Octal)	Valid addresses for Allen Bradley												
10C– 17C	KEEP ALIVE ADDRESS (PC1ALIVE) to (PC8ALIVE)	If ALLENBRD has been entered in Line 7, the Keep Alive Address has a range from 0 to 177717 (octal). Enter the first four most significant digits (1777 maximum) in octal form in lines 10C–17C per the applicable PC. (Range is from 0 to 9999, but only entries in 0 to 1777 range are accepted during a configuration load.) (Default = 0)	The Keep Alive Address is the address within the PC that must be periodically updated by the DHP (or 620 HIM) to inform the PC that the DHP (or 620 HIM) is "alive." If the DHP (or 620 HIM) is "alive," all output data points (if the PC is appropriately configured) are switched to PC ladder logic control. This address is divided into two parts as follows (using address 177717 as an example): <table><tr><td style="text-align: center;">1777</td><td style="text-align: center;">17</td></tr><tr><td style="border: 1px solid black; padding: 5px;">Keep Alive Address</td><td style="border: 1px solid black; padding: 5px;">Keep Alive Bit Position</td></tr></table> The four most significant digits (Keep Alive Address) are entered on this line. The two least significant digits (Keep Alive Bit Position) are entered on Lines 10D–17D per the applicable PC.	1777	17	Keep Alive Address	Keep Alive Bit Position						
1777	17												
Keep Alive Address	Keep Alive Bit Position												

(Continued)

DHP/620-HIM BOX DATA POINT CONFIGURATION FORM
(Continued)

HG DHP BOX POINT DISPL				
25	SLOT 8 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COU
26	SLOT 9 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COUN
27	SLOT 10 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COU
28	SLOT 11 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COUN
29	SLOT 12 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COU
30	SLOT 13 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COUN
31	SLOT 14 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COU
32	SLOT 15 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COUN
33	SLOT 17 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COUN
34	SLOT 18 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COU
35	SLOT 19 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COUN
36	SLOT 20 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COU
37	SLOT 21 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COUN
38	SLOT 22 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COU
39	SLOT 23 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COUN
40	SLOT 24 TYPE	NONE	DIGIN	DIGOUT (Default =
		ANALOGIN	ANALOGOT	COU

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-4 — Instructions for DHP/620-HIM Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG DHP BOX POINT DISPLAY			
		If MODICON or HONYWELL had been entered in Line 7, the Keep Alive Address has a range from 0 to 09999 (Decimal). Enter the four least significant digits (for HONEYWELL or Modicon Model 584) or three least significant digits (for Modicon Models 484 and 384 in decimal form in lines 10C–17C.	This address is divided into two parts as follows (using address 49999 as an example): 0 9999 Specifier Keep Alive Address The four least significant digits (Keep Alive Address) are entered in Lines 10C–17C per the applicable PC. The most significant digit (Specifier, which is always 0) is already entered as a default in Lines 10E–17E.
10D–17D	KEEP ALIVE BIT POSITION (PC1ALVBT) to (PC8ALVBT)	Enter an octal number from 00–17. (Default = 0)	Make entries in Lines 10D–15D only if ALLENBRD is entered in Line 7.
10E–17E	SPECIFIER (PC1ALVSP) to (PC8ALVSP)	(Default = 0)	No entry is required. If MODICON or HONYWELL has been selected in Line 7, only value 0 is legal.
18–32 & 33–47	SLOT 1–15 TYPE (PIUCRDY) SLOT 17–31 TYPE (PIUCRDY) (Default = NONE)	Circle one: NONE DIGIN DIGOUT ANALOGIN ANALOGOT COUNTER	Define the types of data points in DHP slots 1–15 and DHP/620-HIM slots 1–15 and 17–31.

(Continued)

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DHP/620-HIM BOX DATA POINT CONFIGURATION FORM
(Continued)

HG DHP BOX POINT DISPLAY

41	SLOT 25 TYPE	NONE	DIGIN	DIGOUT	
		ANALOGIN	ANALOGOT	COUNTER	(Default = NONE)
42	SLOT 26 TYPE	NONE	DIGIN	DIGOUT	
		ANALOGIN	ANALOGOT	COUNTER	(Default = NONE)
43	SLOT 27 TYPE	NONE	DIGIN	DIGOUT	
		ANALOGIN	ANALOGOT	COUNTER	(Default = NONE)
44	SLOT 28 TYPE	NONE	DIGIN	DIGOUT	
		ANALOGIN	ANALOGOT	COUNTER	(Default = NONE)
45	SLOT 29 TYPE	NONE	DIGIN	DIGOUT	
		ANALOGIN	ANALOGOT	COUNTER	(Default = NONE)
46	SLOT 30 TYPE	NONE	DIGIN	DIGOUT	
		ANALOGIN	ANALOGOT	COUNTER	(Default = NONE)
47	SLOT 31 TYPE	NONE	DIGIN	DIGOUT	
		ANALOGIN	ANALOGOT	COUNTER	(Default = NONE)
48	CHANGE DETECTION (CHNGFLAG)	NOTCONFIG	DETECT	SET	(Default = NOTCONFIG)
49	EVENT PROCESSING (EVENTPRC)	DISABLE	ENABLE		(Default = DISABLE)
50	VARIABLE DEADBAND (BOXVDB)	OFF	ON		(Default = OFF)
60	TOG INTERVAL 1				(Default = 0 seconds)
***	(BOXTOG1)				
61	TOG INTERVAL 2				(Default = 0 seconds)
***	(BOXTOG2)				
61A	SLOT 1 TOG INTERVAL (TOGINTSL)	INTERVL1	INTERVL2	(Default = INTERVL1)	
61B	SLOT 2 TOG INTERVAL	INTERVL1	INTERVL2	(Default = INTERVL1)	
61C	SLOT 3 TOG INTERVAL	INTERVL1	INTERVL2	(Default = INTERVL1)	
61D	SLOT 4 TOG INTERVAL	INTERVL1	INTERVL2	(Default = INTERVL1)	
61E	SLOT 5 TOG INTERVAL	INTERVL1	INTERVL2	(Default = INTERVL1)	
61F	SLOT 6 TOG INTERVAL	INTERVL1	INTERVL2	(Default = INTERVL1)	
61G	SLOT 7 TOG INTERVAL	INTERVL1	INTERVL2	(Default = INTERVL1)	
61H	SLOT 8 TOG INTERVAL	INTERVL1	INTERVL2	(Default = INTERVL1)	

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-4 — Instructions for DHP/620-HIM Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG DHP BOX POINT DISPLAY			
48	CHANGE DETECTION (CHNGFLAG)	Circle one: NOTCONFIG DETECT SET (Default = NOTCONFIG)	If REGULAR is entered in Line 6, make entries in Lines 18–32 for slots 1–15. If EXTENDED is entered in Line 6, make entries in Lines 18–47 for slots 1–15 and 17–31. Slot 16 is reserved for DHP/PC/620 HIM configuration and box status. Slot 32 is reserved for future use. NONE = This slot is not used. DIGIN = Digital Input Data Points. DIGOUT = Digital Output Data Points. ANALOGIN = Analog Input Data Points. ANALOGOT = Analog Output Data Points. COUNTER = Counter Data Points. Select: –NOTCONFIG if no preferred devices are on this Data Hiway. (Speeds up communication.) –DETECT to enable HG to alert LCN devices such as Universal Stations to changes in DHP (or 620 HIM) status that originated in preferred devices on the same Data Hiway. –SET to enable LCN device such as a Universal Station to alert preferred devices to changes it makes in the DHP (or 620 HIM) on the same Data Hiway.

(Continued)

DHP/620-HIM BOX DATA POINT CONFIGURATION FORM
(Continued)

HG DHP BOX POINT DISPLA

52I SLOT 9 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52J SLOT 10 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52K SLOT 11 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52L SLOT 12 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52M SLOT 13 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52N SLOT 14 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52O SLOT 15 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52AA SLOT 17 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52AB SLOT 18 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52AC SLOT 19 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52AD SLOT 20 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52AE SLOT 21 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52AF SLOT 22 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52AG SLOT 23 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52AH SLOT 24 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52AI SLOT 25 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52AJ SLOT 26 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52AK SLOT 27 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52AL SLOT 28 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52AM SLOT 29 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE
52AN SLOT 30 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTER
52A0 SLOT 31 TOG INTERVAL	INTERVL1	INTERVL2 (Default = INTE

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-4 — Instructions for DHP/620-HIM Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG DHP BOX POINT DISPLAY			
49	EVENT PROCESSING (EVENTPRC)	Circle one: DISABLE ENABLE (Default = DISABLE)	Select ENABLE if process alarms, process changes, and operator messages are to be routed through this HG to an LCN.
50 ***	TOG INTERVAL 1 (BOXTOG1)	Enter 0 to 9999 seconds	TOG (Time-Out Gate) intervals are user-specified, selectable maximum times between successive transmissions from a supervisory controller to a particular slot in this DHP (or 620 HIM) box. If the selected interval is exceeded, the slot sheds from its normal mode. If TOG1 or TOG2 is zero, no selection is provided and lines 51A–51AO are skipped.
51 ***	TOG INTERVAL 2 (BOXTOG2)	0 = Never shed (Default = 0 seconds)	
51A–51O	SLOT 1–15 TOG INTERVAL (TOGINTSL)	Circle one: INTERVL1 INTERVL2 (Default = INTERVL1)	Specifies whether TOG INTERVAL 1 or (INTERVL1) TOG INTERVAL 2 (INTERVL2) (lines 50, 51) is to be used as the TOG interval for a given slot.
51AA–51AO	SLOT 17–31 TOG INTERVAL (TOGINTSL)	Circle one: INTERVL1 INTERVL2 (Default = INTERVL1)	Specifies whether TOG INTERVAL 1 (INTERVL1) or TOG INTERVAL 2 (INTERVL2) (lines 51, 52) is to be used as the TOG interval for a given slot.

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EC BOX DATA POINT CONFIGURATION FORM

HG EC BOX POINT DISPL

1	HIWAY NUMBER		(Default = 1
	(HWYNUM		
2	BOX NUMBER		(Default = 5
	(BOXNUM		
3	BOX TYPE	CB AU HLPIU	
	(BOXTYPE)	LEPIU CBRCD EC	
		MC MCRCD DHP	
		H4500 HG GPCI NOTCONFIG (	
EC)			
4	BOX ASSIGNMENT	THISHG REMOTEHG ADDEDHG	(Defa
**	(BOXASSN)		TH
5	LOAD DESTINATION	HG HG HIWAY	(Default =
**	(LOADDEST		
6	CHANGE DETECTION	NOTCONFIG DETECT SET	(Default = NOTC
	(CHNGFLAG		
7	EVENT PROCESSING	DISABLE ENABLE	(Default = DISA
	(EVENTPRC		
8	TOG INTERVAL 1		(Default = 0
***	(BOXTOG1		
9	TOG INTERVAL 2		(Default = 0
***	(BOXTOG2		
9A	SLOT 1 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
	(TOGINTSL		
9B	SLOT 2 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9C	SLOT 3 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9D	SLOT 4 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9E	SLOT 5 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9F	SLOT 6 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9G	SLOT 7 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9H	SLOT 8 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9I	SLOT 9 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9J	SLOT 10 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9K	SLOT 11 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9L	SLOT 12 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9M	SLOT 13 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9N	SLOT 14 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9O	SLOT 15 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9P	SLOT 16 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9Q	SLOT 16 TOG INTERVAL	INTERV1 INTERV2	(Default = INT

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.5 EC Box Data Point (Box/Slot) Configuration Form SW88-534

Table 2-5 contains instructions for Configuration Form *SW88-534*.

Table 2-5 — Instructions for EC Box Data Point Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG EC BOX POINT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN node configuration.*
2	BOX NUMBER (BOXNUM)	Enter a number from 1 to 63 Default = 5)	Defines the hiway address of this EC box on the Data Hiway.
3	BOX TYPE (BOXTYPE)	Enter EC	
4	BOX ASSIGNMENT (BOXASSN)	Circle one: THISHG REMOTEHG ADDEDHG H4500 (Default = THISHG)	Defines the function relative to this EC of <u>each</u> HG on the same physical Data Hiway. Refer to Table 2-2 and to heading 2.1.3. If AddedHG is entered, configuration is complete.
5	LOAD DESTINATION (LOADDEST)	Circle one: HG HG_HIWAY	Define the destination of this configuration information. If ThisHG is entered on line 4 and HG HG_HIWAY is entered on line 5, fill in the rest of this form including lines 8–9Q; otherwise, fill in only lines 6 and 7.
6	CHANGE DETECTION (CHNGFLAG)	Circle one: NOTCONFIG DETECT SET (Default = NOTCONFIG)	Select: –NOTCONFIG if no preferred devices are on this Data Hiway (Speeds up communication.) –DETECT to enable HG to alert LCN devices such as Universal Stations to changes in EC status that originated in preferred devices on the same Data Hiway or in the EC (by external mode switching).

(Continued)

*Refer to Network Form SW88-509, Hiway Gateway Node Display

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EC BOX DATA POINT CONFIGURATION FORM

HG EC BOX POINT DISPL

1	HIWAY NUMBER		(Default = 1
	(HWYNUM		
2	BOX NUMBER		(Default = 5
	(BOXNUM		
3	BOX TYPE	CB AU HLPIU	
	(BOXTYPE)	LEPIU CBRCD EC	
		MC MCRCD DHP	
		H4500 HG GPCI NOTCONFIG (	
EC)			
4	BOX ASSIGNMENT	THISHG REMOTEHG ADDEDHG	(Defa
**	(BOXASSN)		TH
5	LOAD DESTINATION	HG HG HIWAY	(Default =
**	(LOADDEST		
6	CHANGE DETECTION	NOTCONFIG DETECT SET	(Default = NOTC
	(CHNGFLAG		
7	EVENT PROCESSING	DISABLE ENABLE	(Default = DISA
	(EVENTPRC		
8	TOG INTERVAL 1		(Default = 0
***	(BOXTOG1		
9	TOG INTERVAL 2		(Default = 0
***	(BOXTOG2		
9A	SLOT 1 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
	(TOGINTSL		
9B	SLOT 2 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9C	SLOT 3 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9D	SLOT 4 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9E	SLOT 5 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9F	SLOT 6 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9G	SLOT 7 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9H	SLOT 8 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9I	SLOT 9 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9J	SLOT 10 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9K	SLOT 11 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9L	SLOT 12 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9M	SLOT 13 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9N	SLOT 14 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9O	SLOT 15 TOG INTERVAL	INTERV1 INTERV2	(Default = INTE
9P	SLOT 16 TOG INTERVAL	INTERV1 INTERV2	(Default = INT
9Q	SLOT 16 TOG INTERVAL	INTERV1 INTERV2	(Default = INT

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-5 — Instructions for EC Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG EC BOX POINT DISPLAY			
7	EVENT PROCESSING (EVENTPROC)	Circle one: DISABLE ENABLE (Default = DISABLE) Enter 0 to	–SET to enable LCN device, such as a Universal Station, to alert preferred devices to changes it makes in the EC on the same Data Hiway. Select ENABLE if process alarms, process changes, and operator messages are to be routed through this HG to an LCN.
8 ***	TOG INTERVAL 1 (BOXTOG1)	9999 seconds	TOG (Time-Out Gate) intervals are user-specified, selectable maximum times between successive transmissions from a supervisory controller to a particular slot in this EC box. If the selected interval is exceeded, the slot sheds from its normal mode. If TOG1 or TOG2 is zero, no selection is provided and lines 9A–9Q are skipped.
9 ***	TOG INTERVAL 2 (BOXTOG2)	0 = Never shed (Default = 0 seconds)	
(See Note)			
9A–9I	SLOT 1–8 TOG INTERVAL (TOGINTSL)	Circle one: INTERV1 INTERV2 (Default = INTERV1)	Specifies whether TOG INTERVAL 1 (INTERV1) or TOG INTERVAL 2 (INTERV2) entered on Lines 8 and 9 is to be the time-out gate interval for the particular slot.
9J–9Q	SLOT 9–16 TOG INTERVAL (TOGINTSL)	Circle one: INTERV1 INTERV2 (Default = INTERV1)	Specifies whether TOG INTERVAL 1 (INTERV1) or TOG INTERVAL 2 (INTERV2) entered on Lines 8 and 9 is to be the time-out gate interval for the particular slot.

NOTE

In the following instructions, similar entries are grouped to avoid cumbersome repetition. The line numbers of instructions and form entries correspond to each other, but are not shown on facing pages.

HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPLA					
1	HIWAY NUMBER		(Default = 1)		
	(HWYNUM				
2	BOX NUMBER		(Default = 0)		
	(BOXNUM				
3	BOX TYPE	CB	AU	HLPIU	LL
	(BOXTYPE)	LEPIU	CBRCD	EC	E
		MC	MCRCD	DHP	OP
		H4500	HG	GPCI	NOTCONFIG (E
					HLPIU)
4	BOX ASSIGNMENT	THISHG	REMOTEHG	ADDEDHG	SET (Defau
**	(BOXASSN)				(THI
5	LOAD DESTINATION	HG	HG HIWAY	(Default =	
**	(LOADDEST				
6	PIU BOX SIZE	REGULAR	EXTENDED	(Default = REGUL	
	(BOXSIZE				
7	SLOT 1 CARD TYPE	NONE	DIGIN	DIGOUT	(Default = N
	(PIUCRDY)	ANALOGIN	ANALOGOT	COUN	
7A	SLOT 1 CARD OPTION	NOTIFYST	CHNGDECT	SOE	(Default = NOTIF
	(PIUCRDOP				
7B	SLOT 1 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE	
	(PIUCRDOP				
7C	SLOT 1 CARD OPTION	COUNTR16	COUNTR32	(Default = COUNT	
	(PIUCRDOP				
8	SLOT 2 CARD TYPE	NONE	DIGIN	DIGOUT	(Default = N
	(PIUCRDY)	ANALOGIN	ANALOGOT	COUN	
8A	SLOT 2 CARD OPTION	NOTIFYST	CHNGDECT	SOE	(Default = NOTIF
	(PIUCRDOP				
8B	SLOT 2 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE	
	(PIUCRDOP				
8C	SLOT 2 CARD OPTION	COUNTR16	COUNTR32	(Default = COUNT	
	(PIUCRDOP				

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.6 HLPIU Box Data Point (Box/Slot) Configuration Form SW88-535

Table 2-6 contains instructions for Configuration Form *SW88-535*

Table 2-6 — Instructions for HLPIU Box Data Point Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG HLPIU BOX POINT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN node configuration.*
2	BOX NUMBER (BOXNUM)	Enter a number from 1 to 63 (Default = 0)	Defines the hiway address of this HLPIU box on the Data Hiway.
3	BOX TYPE (BOXTYPE)	Enter HLPIU	
4 **	BOX ASSIGNMENT (BOXASSN)	Circle one: THISHG REMOTEHG ADDEDHG (Default = THISHG)	Defines the function relative to this HLPIU of <u>each</u> HG on the same physical Data Hiway. (Refer to Table 2-2 and to heading 2.1.3.) If AddedHG is entered, configuration is complete.
5 **	LOAD DESTINATION (LOADDEST)	Circle one: HG HG_HIWAY (Default = HG)	If ThisHG was entered on line 4 and HG_HIWAY is entered on line 5, fill in the rest of this form, including lines 42–43AF, as applicable; otherwise, fill in the rest of this form except for lines 42–43AF.
6	PIU BOX SIZE (BOXSIZE)	Circle one: REGULAR EXTENDED (Default = REGULAR)	If REGULAR (up-to-16 implemented board slots) is entered, only Lines 7 through 22 are revealed. If EXTENDED (more than 16 implemented board slots) is entered, Lines 23 through 38C are also revealed.

(Continued)

*Refer to Network Form SW88-509, Hiway Gateway Node Display

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HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPL				
9	SLOT 3 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
9A	SLOT 3 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
9B	SLOT 3 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
9C	SLOT 3 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
10	SLOT 4 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
10A	SLOT 4 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
10B	SLOT 4 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
10C	SLOT 4 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
11	SLOT 5 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
11A	SLOT 5 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
11B	SLOT 5 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
11C	SLOT 5 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
12	SLOT 6 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
12A	SLOT 6 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
12B	SLOT 6 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
12C	SLOT 6 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-6 — Instructions for HLPIU Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG HLPIU BOX POINT DISPLAY			
7–38 (See NOTE)	SLOTS 1–32 CARD TYPE (PIUCRDY)	Circle one: NONE DIGIN DIGOUT COUNTER ANALOGOT ANALOGIN (Default = NONE)	NOTE: The following subslot addresses are not available for an HLPUI when analog input cards are installed in these slots: slot 16 subslot 8, and slot 32 subslot 8. Defines a card type for each of the 32 slots in the HLPIU. If NONE, COUNTER, ANALOGOT, or ANALOGIN is entered for a particular slot, skip the SLOT CARD OPTION Lines for that slot and go to the next SLOT CARD TYPE Line for the next slot. NONE = No card in this slot DIGIN = Digital Input DIGOUT = Digital Output COUNTER = Counter Input ANALOGOT = Analog Output ANALOGIN = Analog Input
7A–38A	SLOTS 1–32 CARD OPTION (PIUCRDOP)	If DIGIN has been entered on the SLOT CARD TYPE Line Circle one: NOTIFYST CHNGDECT SOE (Default = No Options)	NOTIFYST = Notify Status CHNGDECT = Change Detect (Latched) SOE = Sequence of Events

(Continued)

NOTE

In the following instructions, similar entries are grouped to avoid cumbersome repetition. The line numbers of instructions and form entries correspond to each other, but are not shown on facing pages.

HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPL				
13	SLOT 7 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
13A	SLOT 7 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
13B	SLOT 7 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
13C	SLOT 7 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
14	SLOT 8 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
14A	SLOT 8 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
14B	SLOT 8 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
14C	SLOT 8 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
15	SLOT 9 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
15A	SLOT 9 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
15B	SLOT 9 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PI9CRDOP			
15C	SLOT 9 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
16	SLOT 10 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
16A	SLOT 10 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
16B	SLOT 10 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
16C	SLOT 10 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-6 — Instructions for HLPIU Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG HLPIU BOX POINT DISPLAY			
7B– 38B	SLOTS 1–32 CARD OPTION (PIUCRDOP)	If DIGOUT has been entered on the SLOT CARD TYPE Line Circle one: PULSEOUT LATCHOUT (Default = No Options)	PULSEOUT = Pulsed Output LATCHOUT = Latched Output
7C– 38C	SLOTS 1–32 CARD OPTION (PIUCRDOP)	If COUNTER has been en- tered on the SLOT CARD TYPE Line Circle one: COUNTR16 COUNTR32 (Default = No Options)	COUNTR16 = 16-bit pulse counter COUNTR32 = 32-bit pulse counter
39	CHANGE DETECTION (CHNGFLAG)	Circle one: NOTCONFG DETECT SET (Default = NOTCONFG)	Select: –NOTCONFG if no preferred devices are on this Data Hiway.(Speeds up commun- ication.) –DETECT to enable HG to alert LCN devices, such as Univer- sal Stations, to changes in HLPIU status that originated in preferred devices on the same Data Hiway. –SET to enable LCN device, such as a Universal Station, to alert preferred to changes it makes in an HLPIU on the same Hiway.
40	EVENT PROCESSING (EVENTPRC)	Circle one: DISABLE ENABLE	Select ENABLE if process alarms, process changes, and operator messages are to be routed through this HG to an LCN.

(Continued)

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HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPL				
17	SLOT 11 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDTY)	ANALOGIN	ANALOGOT	COU
17A	SLOT 11 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
17B	SLOT 11 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
17C	SLOT 11 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
18	SLOT 12 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDTY)	ANALOGIN	ANALOGOT	COU
18A	SLOT 12 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
18B	SLOT 12 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
18C	SLOT 12 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
19	SLOT 13 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDTY)	ANALOGIN	ANALOGOT	COU
19A	SLOT 13 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
19B	SLOT 13 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
19C	SLOT 13 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
20	SLOT 14 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDTY)	ANALOGIN	ANALOGOT	COU
20A	SLOT 14 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
20B	SLOT 14 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
20C	SLOT 14 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-6 — Instructions for HLPIU Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG HLPIU BOX POINT DISPLAY			
41 ***	TOG INTERVAL 1 (BOXTOG1)	Enter 0 to 9999 seconds 0 = Never shed	Time-Out-Gate (TOG) intervals 1 & 2 enable selection of either of two time-out intervals for each slot in this EC.box. The time-out interval is the time between two successive transmissions from a supervisory controller to a particular slot in this box. If a time-out interval is exceeded, the respective slot is automatically placed in the manual mode. If BOXTOG1 OR BOXTOG2 is zero, no selection is provided; skip lines 42A–42AF.
42 ***	TOG INTERVAL 2 (BOXTOG2)	(Default = 0 seconds)	
(See Note)			
42A– 42AF	SLOTS 1–32 TOG INTERVAL (TOGINTSL)	Circle one: INTERV1 INTERV2 (Default = INTERV1)	Specifies whether TOG INTERVAL 1 (INTERV1) or TOG INTERVAL 2 (INTERV2) entered on Lines 41 and 42 is to be the time-out gate interval for the particular slot. Lines 42P–42AF are revealed only if EXTENDED is selected for PIU BOX SIZE on Line 4.

HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPL				
21	SLOT 15 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
21A	SLOT 15 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
21B	SLOT 15 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
21C	SLOT 15 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
22	SLOT 16 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
22A	SLOT 16 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
22B	SLOT 16 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
22C	SLOT 16 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
23	SLOT 17 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
23A	SLOT 17 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
23B	SLOT 17 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
23C	SLOT 17 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
24	SLOT 18 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
24A	SLOT 18 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
24B	SLOT 18 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
24C	SLOT 18 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

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HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPLAY

25	SLOT 19 CARD TYPE	NONE	DIGIN	DIGOUT	(Default = NONE)
	(PIUCRDY)	ANALOGIN	ANALOGOT	COUNTER	
25A	SLOT 19 CARD OPTION	NOTIFYST	CHNGDECT	SOE	(Default = NOTIFYST)
	(PIUCRDOP)				
25B	SLOT 19 CARD OPTION	PULSEOUT	LATCHOUT		(Default = PULSEOUT)
	(PIUCRDOP)				
25C	SLOT 19 CARD OPTION	COUNTR16	COUNTR32		(Default = COUNTR16)
	(PIUCRDOP)				

26	SLOT 20 CARD TYPE	NONE	DIGIN	DIGOUT	(Default = NONE)
	(PIUCRDY)	ANALOGIN	ANALOGOT	COUNTER	
26A	SLOT 20 CARD OPTION	NOTIFYST	CHNGDECT	SOE	(Default = NOTIFYST)
	(PIUCRDOP)				
26B	SLOT 20 CARD OPTION	PULSEOUT	LATCHOUT		(Default = PULSEOUT)
	(PIUCRDOP)				
26C	SLOT 20 CARD OPTION	COUNTR16	COUNTR32		(Default = COUNTR16)
	(PIUCRDOP)				

27	SLOT 21 CARD TYPE	NONE	DIGIN	DIGOUT	(Default = NONE)
	(PIUCRDY)	ANALOGIN	ANALOGOT	COUNTER	
27A	SLOT 21 CARD OPTION	NOTIFYST	CHNGDECT	SOE	(Default = NOTIFYST)
	(PIUCRDOP)				
27B	SLOT 21 CARD OPTION	PULSEOUT	LATCHOUT		(Default = PULSEOUT)
	(PIUCRDOP)				
27C	SLOT 21 CARD OPTION	COUNTR16	COUNTR32		(Default = COUNTR16)
	(PIUCRDOP)				

28	SLOT 22 CARD TYPE	NONE	DIGIN	DIGOUT	(Default = NONE)
	(PIUCRDY)	ANALOGIN	ANALOGOT	COUNTER	
28A	SLOT 22 CARD OPTION	NOTIFYST	CHNGDECT	SOE	(Default = NOTIFYST)
	(PIUCRDOP)				
28B	SLOT 22 CARD OPTION	PULSEOUT	LATCHOUT		(Default = PULSEOUT)
	(PIUCRDOP)				
28C	SLOT 22 CARD OPTION	COUNTR16	COUNTR32		(Default = COUNTR16)
	(PIUCRDOP)				

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPL				
29	SLOT 23 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
29A	SLOT 23 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
29B	SLOT 23 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
29C	SLOT 23 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
30	SLOT 24 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
30A	SLOT 24 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
30B	SLOT 24 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
30C	SLOT 24 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
31	SLOT 25 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
31A	SLOT 25 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
31B	SLOT 25 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
31C	SLOT 25 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
32	SLOT 26 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
32A	SLOT 26 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
32B	SLOT 26 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
32C	SLOT 26 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

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HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPL				
33	SLOT 27 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
33A	SLOT 27 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
33B	SLOT 27 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
33C	SLOT 27 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
34	SLOT 28 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
34A	SLOT 28 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
34B	SLOT 28 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
34C	SLOT 28 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
35	SLOT 29 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
35A	SLOT 29 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
35B	SLOT 29 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
35C	SLOT 29 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
36	SLOT 30 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
36A	SLOT 30 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
36B	SLOT 30 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
36C	SLOT 30 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPL				
37	SLOT 31 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
37A	SLOT 31 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
37B	SLOT 31 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
37C	SLOT 31 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			
38	SLOT 32 CARD TYPE	NONE	DIGIN	DIGOUT (Default =
	(PIUCRDY)	ANALOGIN	ANALOGOT	COU
38A	SLOT 32 CARD OPTION	NOTIFYST	CHNGDECT	SOE (Default = NOTI
	(PIUCRDOP			
38B	SLOT 32 CARD OPTION	PULSEOUT	LATCHOUT	(Default = PULSE
	(PIUCRDOP			
38C	SLOT 32 CARD OPTION	COUNTR16	COUNTR32	(Default = COUN
	(PIUCRDOP			

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

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HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPL

39	CHANGE DETECTION	NOTCONFIG	DETECT	SET	(Default = NOTC
	(CHNGFLAG				
40	EVENT PROCESSING	DISABLE	ENABLE		(Default = DIS
	(EVENTPRC				
41	TOG INTERVAL 1				(Default = 0 seconds
***	(BOXTOG1				
42	TOG INTERVAL 2				(Default = 0 seconds
***	(BOXTOG2				
42A	SLOT 1 TOG INTERVAL	INTERV1	INTERV2		(Default = INTE
	(TOGINTSL				
42B	SLOT 2 TOG INTERVAL	INTERV1	INTERV2		(Default = INT
42C	SLOT 3 TOG INTERVAL	INTERV1	INTERV2		(Default = INTE
42D	SLOT 4 TOG INTERVAL	INTERV1	INTERV2		(Default = INT
42E	SLOT 5 TOG INTERVAL	INTERV1	INTERV2		(Default = INTE
42F	SLOT 6 TOG INTERVAL	INTERV1	INTERV2		(Default = INT
42G	SLOT 7 TOG INTERVAL	INTERV1	INTERV2		(Default = INTE
42H	SLOT 8 TOG INTERVAL	INTERV1	INTERV2		(Default = INT
42I	SLOT 9 TOG INTERVAL	INTERV1	INTERV2		(Default = INTE
42J	SLOT 10 TOG INTERVAL	INTERV1	INTERV2		(Default = INT
42K	SLOT 11 TOG INTERVAL	INTERV1	INTERV2		(Default = INTE
42L	SLOT 12 TOG INTERVAL	INTERV1	INTERV2		(Default = INT
42M	SLOT 13 TOG INTERVAL	INTERV1	INTERV2		(Default = INTE
42N	SLOT 14 TOG INTERVAL	INTERV1	INTERV2		(Default = INT
42O	SLOT 15 TOG INTERVAL	INTERV1	INTERV2		(Default = INTE

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

HLPIU BOX DATA POINT CONFIGURATION FORM

HG HLPIU BOX POINT DISPL						
42P	SLOT 16	TOG	INTERVAL	INTERV1	INTERV2	(Default = INT
42Q	SLOT 17	TOG	INTERVAL	INTERV1	INTERV2	(Default = INTE
42R	SLOT 18	TOG	INTERVAL	INTERV1	INTERV2	(Default = INT
42S	SLOT 19	TOG	INTERVAL	INTERV1	INTERV2	(Default = INTE
42T	SLOT 20	TOG	INTERVAL	INTERV1	INTERV2	(Default = INT
42U	SLOT 21	TOG	INTERVAL	INTERV1	INTERV2	(Default = INTE
42V	SLOT 22	TOG	INTERVAL	INTERV1	INTERV2	(Default = INT
42W	SLOT 23	TOG	INTERVAL	INTERV1	INTERV2	(Default = INTE
42X	SLOT 24	TOG	INTERVAL	INTERV1	INTERV2	(Default = INT
42Y	SLOT 25	TOG	INTERVAL	INTERV1	INTERV2	(Default = INTE
42Z	SLOT 26	TOG	INTERVAL	INTERV1	INTERV2	(Default = INT
42AA	SLOT 27	TOG	INTERVAL	INTERV1	INTERV2	(Default = INTE
42AB	SLOT 28	TOG	INTERVAL	INTERV1	INTERV2	(Default = INT
42AC	SLOT 29	TOG	INTERVAL	INTERV1	INTERV2	(Default = INTE
42AD	SLOT 30	TOG	INTERVAL	INTERV1	INTERV2	(Default = INT
42AE	SLOT 31	TOG	INTERVAL	INTERV1	INTERV2	(Default = INTE
42AF	SLOT 32	TOG	INTERVAL	INTERV1	INTERV2	(Default = INT

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

LEPIU BOX DATA POINT CONFIGURATION FORM

HG LEPIU BOX POINT DISPLA					
1	HIWAY NUMBER		(Default = 1		
	(HWYNUM				
2	BOX NUMBER		(Default = 5		
	(BOXNUM				
3	BOX TYPE	CB	AU	HLPIU	L
	(BOXTYPE)	LEPIU	CBRCD	EC	E
		MC	MCRCD	DHP	O
		H4500	HG	GPCI	NOTCONF (
	LEPIU)				
4	BOX ASSIGNMENT	THISHG	REMOTEHG	ADDEDHG	H
	(BOXASSN)				(Default = TH
5	LOAD DESTINATION	HG	HG HIWAY	(Default =	
	(LOADDEST				
6	PIU BOX SIZE	REGULAR	EXTENDED	(Default = REG	
	(BOXSIZE				
7	SLOT 1 CARD TYPE	NONE	ANALOGIN	(Default = NONE for	
	(PIUCRDY)			Card S	
8	SLOT 2 CARD TYPE	NONE	ANALO		
9	SLOT 3 CARD TYPE	NONE	ANALO		
10	SLOT 4 CARD TYPE	NONE	ANALO		
11	SLOT 5 CARD TYPE	NONE	ANALO		
12	SLOT 6 CARD TYPE	NONE	ANALO		
13	SLOT 7 CARD TYPE	NONE	ANALO		
14	SLOT 8 CARD TYPE	NONE	ANALO		
15	SLOT 9 CARD TYPE	NONE	ANALO		
16	SLOT 10 CARD TYPE	NONE	ANALO		
17	SLOT 11 CARD TYPE	NONE	ANALO		
18	SLOT 12 CARD TYPE	NONE	ANALO		
19	SLOT 13 CARD TYPE	NONE	ANALO		
20	SLOT 14 CARD TYPE	NONE	ANALO		
21	SLOT 15 CARD TYPE	NONE	ANALO		
22	SLOT 16 CARD TYPE	NONE	ANALO		

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.7 LEPIU Box Data Point (Box/Slot) Configuration Form SW88-536

Table 2-7 contains instructions for Configuration Form *SW88-536*.

Table 2-7— Instructions for LEPIU Box Data Point Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG LEPIU BOX POINT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN node configuration.*
2	BOX NUMBER (BOXNUM)	Enter a number from 1 to 63 (Default = 0)	Defines the hiway address of this LEPIU box on the Data Hiway.
3	BOX TYPE (BOXTYPE)	Enter LEPIU	
4	BOX ASSIGNMENT (BOXASSN)	Circle one: THISHG REMOTEHG ADDEDHG H4500 (Default = THISHG)	Defines the function relative to this LEPIU of <u>each</u> HG on the same physical Data Hiway. (Refer to Table 2-2 and to heading 2.1.3.) If AddedHG is entered, configuration is complete.
5	LOAD DESTINATION (LOADDEST)	Circle one: HG HG_HIWAY (Default = HG)	Defines the destination of configuration data.
6	PIU BOX SIZE (BOXSIZE)	Circle one: REGULAR EXTENDED (Default = REGULAR)	If REGULAR (up-to-16 implemented slots) is entered, make entries in Lines 7 through 22. If EXTENDED (with more than 16 implemented slots) is entered, make entries in Lines 7 through 38.

(Continued)

NOTE

In the following instructions, similar entries are grouped to avoid cumbersome repetition. The line numbers of instructions and form entries correspond to each other, but are not shown on facing pages.

*Refer to Network Form SW88-509, Hiway Gateway Node Display

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LEPIU BOX DATA POINT CONFIGURATION FORM
(Continued)

HG LEPIU BOX POINT DISPL				
23	SLOT 17 CARD TYPE	NONE	ANALOGIN	(Default = NONE for
24	SLOT 18 CARD TYPE	NONE	ANALOGIN	Card S1
25	SLOT 19 CARD TYPE	NONE	ANALO	
26	SLOT 20 CARD TYPE	NONE	ANALO	
27	SLOT 21 CARD TYPE	NONE	ANALO	
28	SLOT 22 CARD TYPE	NONE	ANALO	
29	SLOT 23 CARD TYPE	NONE	ANALO	
30	SLOT 24 CARD TYPE	NONE	ANALO	
31	SLOT 25 CARD TYPE	NONE	ANALO	
32	SLOT 26 CARD TYPE	NONE	ANALO	
33	SLOT 27 CARD TYPE	NONE	ANALO	
34	SLOT 28 CARD TYPE	NONE	ANALO	
35	SLOT 29 CARD TYPE	NONE	ANALO	
36	SLOT 30 CARD TYPE	NONE	ANALO	
37	SLOT 31 CARD TYPE	NONE	ANALO	
38	SLOT 32 CARD TYPE	NONE	ANALO	
39	CHANGE DETECTION	NOTCONFIG	DETECT	SET (Default = NOTC
	(CHNGFLAG			
40	EVENT PROCESSING	DISABLE	ENABLE	(Default = DIS
	(EVENTPRC			

Table 2-7 — Instructions for LEPIU Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG LEPIU BOX POINT DISPLAY			
7–38 (See NOTE)	SLOTS 1–32 CARD TYPE (PIUCRDY)	Circle one: NONE ANALOGIN (Default = NONE)	NOTE: The following subslot addresses are not available for use in the LEPIU: slot 16 subslot 8 (Remote box 8 last point), and slot 32 subslot 8 (Remote box16 last point). Defines a card type for each of the 32 slots in the LEPIU. NONE = No card in this slot ANALOGIN Analog Input
39	CHANGE DETECTION (CHNGFLAG)	Circle one: NOTCONFIG DETECT SET (Default = NOTCONFIG)	Select: –NOTCONFIG if no preferred devices are on this Data Hiway. (Speeds up communication.) –DETECT to enable HG to alert LCN devices, such as Universal Stations, to changes in LEPIU status that originated in preferred devices on the same Data Hiway. –SET to enable LCN device, such as a Universal Station, to alert preferred devices to changes it makes to an LEPIU on the same Data Hiway.
40	EVENT PROCESSING (EVENTPRC)	Circle one: DISABLE ENABLE (Default = DISABLE)	Select ENABLE if process alarms, process changes, and operator messages are to be routed through this HG to an LCN.

LLPIU BOX DATA POINT CONFIGURATION FORM

HG LLPIU BOX POINT DISPLA					
1	HIWAY NUMBER		(Default = 1		
	(HWYNUM				
2	BOX NUMBER		(Default = 5		
	(BOXNUM				
3	BOX TYPE	CB	AU	HLPIU	L
	(BOXTYPE)	LEPIU	CBRC	EC	E
		MC	MCRCD	DHP	O
		H4500	HG	GPCI	NOTCONF (Enter
4	BOX ASSIGNMENT	THISHG	REMOTEHG	ADDEDHG	H
	(BOXASSN)	(Default = THIS			
5	LOAD DESTINATION	HG	HG HIWAY	(Default =	
	(LOADDEST				
6	PIU BOX SIZE	REGULAR	EXTEN		
	(BOXSIZE				
7	SLOT 1 CARD TYPE	NONE	ANALOGIN	(Default = NONE fo	
	(PIUCRDY)			Card T	
8	SLOT 2 CARD TYPE	NONE	ANALO		
9	SLOT 3 CARD TYPE	NONE	ANALO		
10	SLOT 4 CARD TYPE	NONE	ANALO		
11	SLOT 5 CARD TYPE	NONE	ANALO		
12	SLOT 6 CARD TYPE	NONE	ANALO		
13	SLOT 7 CARD TYPE	NONE	ANALO		
14	SLOT 8 CARD TYPE	NONE	ANALO		
15	SLOT 9 CARD TYPE	NONE	ANALO		
16	SLOT 10 CARD TYPE	NONE	ANALO		
17	SLOT 11 CARD TYPE	NONE	ANALO		
18	SLOT 12 CARD TYPE	NONE	ANALO		
19	SLOT 13 CARD TYPE	NONE	ANALO		
20	SLOT 14 CARD TYPE	NONE	ANALO		
21	SLOT 15 CARD TYPE	NONE	ANALO		
22	SLOT 16 CARD TYPE	NONE	ANALO		

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.8 LLPIU Box Data Point (Box/Slot) Configuration Form SW88-537

Table 2-8 contains instructions for Configuration Form *SW88-537*.

Table 2-8 — Instructions for LLPIU Box Data Point Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG LLPIU BOX POINT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN node configuration*.
2	BOX NUMBER (BOXNUM)	Enter a number from 1 to 63 (Default = 0)	Defines the hiway address of this LLPIU box on the Data Hiway.
3	BOX TYPE (BOXTYPE)	Enter LLPIU	
4	BOX ASSIGNMENT (BOXASSN)	Circle one: THISHG REMOTEHG ADDEDHG H4500 (Default = THISHG)	Defines the function relative to this LLPIU of <u>each</u> HG on the same physical Data Hiway. (Refer to Table 2-2 and to heading 2.1.3.) If AddedHG is entered, configuration is complete.
5	LOAD DESTINATION (LOADDEST)	Circle one: HG HG_HIWAY (Default = HG)	Defines the destination of configuration data.
6	PIU BOX SIZE (BOXSIZE)	Circle one: REGULATOR EXTENDED (Default = REGULAR)	If REGULAR (up-to-16 implemented slots) is entered, make entries in Lines 7 through 22. If EXTENDED (more than 16 implemented slots) is entered, make entries in Lines 7 through 38.

(Continued)

NOTE

In the following instructions, similar entries are grouped to avoid cumbersome repetition. The line numbers of instructions and form entries correspond to each other, but are not shown on facing pages.

*Refer to Network Form SW88-509, Hiway Gateway Node Display

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LLPIU BOX DATA POINT CONFIGURATION FORM
(Continued)

HG LLPIU BOX POINT DISPLAY				
23	SLOT 17 CARD TYPE	NONE	ANALOGIN	(Default = NONE for all Card Types)
24	SLOT 18 CARD TYPE	NONE	ANALOGIN	
25	SLOT 19 CARD TYPE	NONE	ANALOGIN	
26	SLOT 20 CARD TYPE	NONE	ANALOGIN	
27	SLOT 21 CARD TYPE	NONE	ANALOGIN	
28	SLOT 22 CARD TYPE	NONE	ANALOGIN	
29	SLOT 23 CARD TYPE	NONE	ANALOGIN	
30	SLOT 24 CARD TYPE	NONE	ANALOGIN	
31	SLOT 25 CARD TYPE	NONE	ANALOGIN	
32	SLOT 26 CARD TYPE	NONE	ANALOGIN	
33	SLOT 27 CARD TYPE	NONE	ANALOGIN	
34	SLOT 28 CARD TYPE	NONE	ANALOGIN	
35	SLOT 29 CARD TYPE	NONE	ANALOGIN	
36	SLOT 30 CARD TYPE	NONE	ANALOGIN	
37	SLOT 31 CARD TYPE	NONE	ANALOGIN	
38	SLOT 32 CARD TYPE	NONE	ANALOGIN	
39	CHANGE DETECTION (CHNGFLAG)	NOTCONFIG	DETECT	SET (Default = NOTCONFIG)
40	EVENT PROCESSING (EVENTPRC)	DISABLE	ENABLE	(Default = DISABLE)
41	VARIABLE DEADBAND (BOXVDB)	OFF	ON	(Default = OFF)

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-8 — Instructions for LLPIU Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG LLPIU BOX POINT DISPLAY			
8.			NOTE: The following subslot addresses are not available for use in an LLPIU: slot 1 subslot 1-4, slot 16 subslot 8, and slot 32 subslot
7-38 (See Note)	SLOTS 1-32 CARD TYPE (PIUCRDY)	Circle one: NONE ANALOGIN (Default = NONE)	Defines the card type for each of the 32 slots in the LLPIU.
39	CHANGE DETECTION (CHNGFLAG)	Circle one: NOTCONFIG DETECT SET (Default = NOTCONFIG)	Select: -NOTCONFIG if no preferred devices are on this Data Hiway. (Speeds up communication.) -DETECT to enable HG to alert LCN devices such as Universal Stations to changes in LLPIU status that originated in preferred devices on the same Data Hiway. -SET to enable LCN device such as a Universal Station to alert preferred devices to changes it makes in an LLPIU on the same Hiway.
40	EVENT PROCESSING (EVENTPRC)	Circle one: DISABLE ENABLE (Default = DISABLE)	Select ENABLE if process alarms, process changes, and operator messages are to be routed through this HG to an LCN.
41	VARIABLE DEADBAND	OFF ON (Default = OFF)	Select ON to prevent PV alarm deadbands from being initialized to 1% during a checkpoint box load request when variable deadband failure is present. Select OFF when variable deadband firmware is first installed so PV alarm deadbands will be initialized during a checkpoint box load.

MC BOX DATA POINT CONFIGURATION FORM

HG MC BOX POINT DISPLA

1	HIWAY NUMBER		(Default = 1
	(HWYNUM		
2	BOX NUMBER		(Default = 0
	(BOXNUM		
3	BOX TYPE	CB AU HLPIU L	
	(BOXTYPE)	LEPIU CBRCD EC E	
		MC MCRCD DHP O	
		H4500 HG GPCI NOTCONFG (Ente	
4	BOX ASSIGNMENT	THISHG REMOTEHG ADDEDHG H	
**	(BOXASSN)	(Default = THIS	
5	LOAD DESTINATION	HG HG HIWAY (Default =	
**	(LOADDEST		
6	MC REVISION CODE	REV20 REV30 REV31 (Default = R	
	(MCREV		
7	BOX TRENDING	NOTREND TREND (Default = NOTR	
	(BOXTREND		
8	ADDITIONAL SOPL MEMORY	NOSOPL SOPL (Default = NOS	
	(SOPLMEM		
9	C-LINK NUMBER		(Default = 1
	(BOXCLINK		
10	ADDRESS NUMBER ON C-L		(Default = 1
	(CLINKNUM		
11	LIBRARY	(Default = 1	
	(LIBRYNUM		
12	NUMBER OF I/O FILES	(Default = 0	
	(NMBRBCFL		
12A	SLOT 1 CARD TYPE	NONE DIGIN DIGOUT ANALOGOT CO	
	(PIUCRDY)	(Default = NO	
12A1	SLOT 1 CARD OPTION	STATUS LATCHIN (Default = neit	
12A2	SLOT 1 CARD OPTION	PULSEOUT LATCHOUT (Default = nei	
12B	SLOT 2 CARD TYPE	NONE DIGIN DIGOUT ANALOGOT CO	
	(Default = NO		
12B1	SLOT 2 CARD OPTION	STATUS LATCHIN (Default = neit	
12B2	SLOT 2 CARD OPTION	PULSEOUT LATCHOUT (Default = nei	
12C	SLOT 3 CARD TYPE	NONE DIGIN DIGOUT ANALOGOT CO	
	(Default = NO		
12C1	SLOT 3 CARD OPTION	STATUS LATCHIN (Default = neit	
12C2	SLOT 3 CARD OPTION	PULSEOUT LATCHOUT (Default = nei	
12D	SLOT 4 CARD TYPE	NONE DIGIN DIGOUT ANALOGOT CO	
	(Default = NO		
12D1	SLOT 4 CARD OPTION	STATUS LATCHIN (Default = neit	
12D2	SLOT 4 CARD OPTION	PULSEOUT LATCHOUT (Default = nei	

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.9 MC Box Data Point (Box/Slot) Configuration Form SW88-538

Table 2-9 contains instructions for Configuration Form *SW88-538*.

Table 2-9 — Instructions for MC Box Data Point Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG MC BOX POINT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN node configuration.*
2	BOX NUMBER (BOXNUM)	Enter a number from 1 to 63 (box number 4 is reserved for the Operator Station.) (Default = 0)	Defines the box address of this MC box on the selected Data Hiway. If a box address of less than 32 is entered, fill out Lines 4 through 8. If the box address of 32 or greater is entered, skip Lines 4 through 8 and go to Line 9.
3	BOX TYPE (BOXTYPE)	Enter MC	Defines this box as being the Multifunction Controller.
4 **	BOX ASSIGNMENT (BOXASSN)	Circle one: THISHG REMOTEHG H4500 ADDEDHG (Default = THISHG)	Defines the function relative to this MC of <u>each</u> HG on the same physical Data Hiway. Refer to Table 3-25 and to heading 2.1.3. If AddedHG is entered, configuration is complete.
5 **	LOAD DESTINATION (LOADDEST)	Circle one: HG HG_HIWAY (Default = HG)	If ThisHG was entered on line 4 and HG HIWAY is entered on line 5, fill in the rest of this form, including lines 15–16Q. Otherwise, fill in the rest of this form, as applicable, except for lines 15–16Q.

(Continued)

*Refer to Network Form SW88-509, Hiway Gateway Node Display

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MC BOX DATA POINT CONFIGURATION FORM
(Continued)

HG MC BOX POINT DISPL					
12E SLOT 5 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12E1 SLOT 5 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12E2 SLOT 5 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12F SLOT 6 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12F1 SLOT 6 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12F2 SLOT 6 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12G SLOT 7 CARD TYPE	NONE	DIGIN	DI		
	ANALOGIN	ANALOGOT	COUNTER	(Default =	
12G1 SLOT 7 CARD OPTION	STATUS	LATCHIN	(Default = neit		
12G2 SLOT 7 CARD OPTION	PULSEOUT	LATCHOUT	(Default = nei		
12H SLOT 8 CARD TYPE	NONE	DIGIN	DIG		
	ANALOGIN	ANALOGOT	COUNTER	(Default =	
12H1 SLOT 8 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12H2 SLOT 8 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12J SLOT 9 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12J1 SLOT 9 CARD OPTION	STATUS	LATCHIN	(Default = neit		
12J2 SLOT 9 CARD OPTION	PULSEOUT	LATCHOUT	(Default = nei		
12K SLOT 10 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12K1 SLOT 10 CARD OPTION	STATUS	LATCHIN	(Default = neit		
12K2 SLOT 10 CARD OPTION	PULSEOUT	LATCHOUT	(Default = nei		
12L SLOT 11 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12L1 SLOT 11 CARD OPTION	STATUS	LATCHIN	(Default = neit		
12L2 SLOT 11 CARD OPTION	PULSEOUT	LATCHOUT	(Default = nei		
12M SLOT 12 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12M1 SLOT 12 CARD OPTION	STATUS	LATCHIN	(Default = neit		
12M2 SLOT 12 CARD OPTION	PULSEOUT	LATCHOUT	(Default = nei		

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-9 — Instructions for MC Box Data Point Configuration Form (Continued)

Table 2-1 Instructions for MC Box Data Point Configuration (Continued)

LINE	NAME	FORM ENTRY	REMARKS								
HG MC BOX POINT DISPLAY											
6	MC REVISION CODE (MCREV)	Circle one: REV20 REV30 REV 31 (Default = REV 30)	REV20—No UAC in MC. REV30—With UAC in MC. REV31—With Normal Mode handling								
7	BOX TRENDING (BOXTREND)	Circle one: NOTREND TREND (Default = NOTREND)	NOTREND—MC box does not have trend option. TREND—MC box has trend option.								
8	ADDITIONAL SOPL MEMORY (SOPLMEM)	Circle one: NOSOPL SOPL (Default = NOSOPL)	Indicates that additional SOPL memory is in this box. Make an entry in this line only if a) REV 20 is entered in Line 6 and NOTREND is entered in Line 7, or b. REV30 or REV31 is entered in line 6.								
9	C-LINK NUMBER (BOXCLINK)	Enter a number from 1 to 8 (Default = 1)	Defines the C-Link number to which this MC is connected.								
10	ADDRESS NUMBER ON C-LINK (CLINKNUM)	Enter a number from 1 to 8 (Default = 1)	Defines the box address number on the C-Link.								
11	LIBRARY (LIBRYNUM)	Enter a number from 1 to 4 (Default = 1)	Defines HG library that is used by this MC. Refer to instructions for HG Library Configuration Form.								
12	NUMBER OF I/O FILES (NMBRBCFL)	Enter 0, 1, or 2 (Default = 0)	Defines the number of I/O files that are attached to this MC box as follows: <table><tr><th>Entry</th><th>Description</th></tr><tr><td>0</td><td>No I/O files</td></tr><tr><td>1</td><td>One I/O file (8 slots)</td></tr><tr><td>2</td><td>Two I/O files (16 slots)</td></tr></table>	Entry	Description	0	No I/O files	1	One I/O file (8 slots)	2	Two I/O files (16 slots)
Entry	Description										
0	No I/O files										
1	One I/O file (8 slots)										
2	Two I/O files (16 slots)										

(Continued)

MC BOX DATA POINT CONFIGURATION FORM
(Continued)

HG MC BOX POINT DISPLAY (CONTINUE)

12N SLOT 13 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12N1 SLOT 13 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12N2 SLOT 13 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12P SLOT 14 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12P1 SLOT 14 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12P2 SLOT 14 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12Q SLOT 15 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12Q1 SLOT 15 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12Q2 SLOT 15 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12R SLOT 16 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12R1 SLOT 16 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12R2 SLOT 16 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
13 CHANGE DETECTION	NOTCONFG	DETECT	SET	(Default = NOTC	
	(CHNGFLAG				
14 EVENT PROCESSING	DISABLE	ENABLE	(Default = DIS		
	(EVENTPRC				
15 TOG INTERVAL 1			(Default = 0 seconds		
*** (BOXTOG1					
16 TOG INTERVAL 2			(Default = 0 seconds		
*** (BOXTOG2					
16A SLOT 1 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
	(TOGINTSL				
16B SLOT 2 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		
16C SLOT 3 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
16D SLOT 4 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		
16E SLOT 5 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
16F SLOT 6 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		
16H SLOT 7 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
16I SLOT 8 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-9 — Instructions for MC Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG MC BOX POINT DISPLAY			
12A– 12H See NOTE)	SLOT 1–8 CARD TYPE (PIUCRDY)	Circle one: NONE DIGIN DIGOUT COUNTER ANALOGOT ANALOGIN (Default = NONE)	If no I/O files are to be configured skip Lines 12A–12R2 (SLOT CARD TYPE and SLOT CARD OPTION) for slots 1–16 and go to Line 13. If one I/O file is to be configured, fill out the SLOT CARD TYPE and SLOT CARD OPTION Lines for slots 1–8 (Lines 12A–12H2). If two I/O files are to be configured, fill out the SLOT CARD TYPE and SLOT CARD OPTION Lines for slots 1–16 (Lines 12A–12R2). Defines a card type for each of the 8 slots (slots 1–8) in I/O File 1 If NONE, COUNTER, ANALOGOT, or ANALOGIN is entered for a particular slot, skip the SLOT CARD OPTION lines for the slot and go on to the next SLOT CARD TYPE line from the next slot. (The ANALOGIN card type can be specified for only slots 7 and 8.)

(Continued)

NOTE

In the following instructions, similar entries are grouped to avoid cumbersome repetition. The line numbers of instructions and form entries correspond to each other, but are not shown on facing pages.

MC BOX DATA POINT CONFIGURATION FORM
(Continued)

HG MC BOX POINT DISPLAY (CONTINUE)

12N SLOT 13 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12N1 SLOT 13 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12N2 SLOT 13 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12P SLOT 14 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12P1 SLOT 14 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12P2 SLOT 14 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12Q SLOT 15 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12Q1 SLOT 15 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12Q2 SLOT 15 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12R SLOT 16 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12R1 SLOT 16 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12R2 SLOT 16 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
13 CHANGE DETECTION	NOTCONF	DETECT	SET	(Default = NOTC	
	(CHNGFLAG				
14 EVENT PROCESSING	DISABLE	ENABLE	(Default = DIS		
	(EVENTPRC				
15 TOG INTERVAL 1			(Default = 0 seconds		
*** (BOXTOG1					
16 TOG INTERVAL 2			(Default = 0 seconds		
*** (BOXTOG2					
16A SLOT 1 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
	(TOGINTSL				
16B SLOT 2 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		
16C SLOT 3 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
16D SLOT 4 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		
16E SLOT 5 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
16F SLOT 6 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		
16H SLOT 7 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
16I SLOT 8 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-9 — Instructions for MC Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG MC BOX POINT DISPLAY			
12A1– 12H1	SLOT 1–8 CARD OPTION (PIUCRDOP)	If DIGIN has been entered on the SLOT CARD TYPE line, circle one: STATUS LATCHIN (Default = No Options)	NONE = No card in this slot DIGIN = Digital Input DIGOUT = Digital Output COUNTER = Counter Input ANALOGOT = Analog Output ANALOGIN = Analog Input (Slots 7 & 8 only) STATUS = Status LATCHIN = Change Detect (Latched)
12A2– 12H2	SLOT 1–8 CARD OPTION (PIUCRDOP)	If DIGOUT has been entered on the SLOT CARD TYPE line, Circle one: PULSEOUT LATCHOUT (Default = No Options)	PULSEOUT = Pulsed Output LATCHOUT = Latched Output
12J– 12R	SLOT 1–16 CARD TYPE (PIUCRDY)	Circle one: NONE DIGIN DIGOUT COUNTER ANALOGOT (Default = NONE)	Defines the card type for each of the 8 slots (slots 9–16) in I/O File 2. NONE = No card in this slot DIGIN = Digital Input DIGOUT = Digital Output COUNTER = Counter Input ANALOGOT = Analog Output

(Continued)

MC BOX DATA POINT CONFIGURATION FORM
(Continued)

HG MC BOX POINT DISPLAY (CONTINUE)

12N SLOT 13 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12N1 SLOT 13 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12N2 SLOT 13 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12P SLOT 14 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12P1 SLOT 14 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12P2 SLOT 14 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12Q SLOT 15 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12Q1 SLOT 15 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12Q2 SLOT 15 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
12R SLOT 16 CARD TYPE	NONE	DIGIN	DIGOUT	ANALOGOT	CO
	(Default = NO				
12R1 SLOT 16 CARD OPTION	STATUS	LATCHIN	(Default = nei		
12R2 SLOT 16 CARD OPTION	PULSEOUT	LATCHOUT	(Default = neit		
13 CHANGE DETECTION	NOTCONF	DETECT	SET	(Default = NOTC	
	(CHNGFLAG				
14 EVENT PROCESSING	DISABLE	ENABLE	(Default = DIS		
	(EVENTPRC				
15 TOG INTERVAL 1			(Default = 0 seconds		
*** (BOXTOG1					
16 TOG INTERVAL 2			(Default = 0 seconds		
*** (BOXTOG2					
16A SLOT 1 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
	(TOGINTSL				
16B SLOT 2 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		
16C SLOT 3 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
16D SLOT 4 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		
16E SLOT 5 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
16F SLOT 6 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		
16H SLOT 7 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE		
16I SLOT 8 TOG INTERVAL	INTERV1	INTERV2	(Default = INT		

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-9 — Instructions for MC Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG MC BOX POINT DISPLAY			
12J1– 12R1	SLOT 9–16 CARD OPTION (PIUCRDOP)	If DIGIN has been entered on the SLOT CARD TYPE line Circle one: STATUS LATCHIN (Default = No Options)	STATUS = Status LATCHIN = Change Detect (Latched)
12J2– 12R2	SLOT 9–16 CARD OPTION (PIUCRDOP)	If DIGOUT has been entered on the SLOT CARD TYPE line Circle one: PULSEOUT LATCHOUT (Default = No Options)	PULSEOUT = Pulsed Output LATCHOUT = Latched Output
13	CHANGE DETECTION (CHNGFLAG)	Circle one: NOTCONFIG DETECT SET (Default = NOTCONFIG)	Select: –DETECT to enable HG to alert LCN devices such as Univer- sal Stations to changes in MC status that originated in pre- ferred devices on the same Data Hiway or in the the Change Circular list of the MC. –SET to enable LCN device such as a Universal Station to alert preferred devices to changes it makes in an MC on the same Hiway.
14	EVENT PROCESSING (EVENTPRC)	Circle one: DISABLE ENABLE (Default = DISABLE)	Select ENABLE if process alarms, process changes, and operator messages are to be routed through this HG to an LCN.

(Continued)

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MC BOX DATA POINT CONFIGURATION FORM
(Continued)

HG MC BOX POINT DISPLAY (CONTINUE)

16J SLOT 9 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
16K SLOT 10 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
16L SLOT 11 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
16M SLOT 12 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
16N SLOT 13 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
16O SLOT 14 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
16P SLOT 15 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
16Q SLOT 16 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
17A IO SLOT 1 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
17B IO SLOT 2 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
17C IO SLOT 3 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
17D IO SLOT 4 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
17E IO SLOT 5 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
17F IO SLOT 6 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
17G IO SLOT 7 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
17H IO SLOT 8 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
17I IO SLOT 9 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
17J IO SLOT 10 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
17K IO SLOT 11 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
17L IO SLOT 12 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
17M IO SLOT 13 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
17N IO SLOT 14 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
17O IO SLOT 15 TOG INTERVAL	INTERV1	INTERV2	(Default = INT
17P IO SLOT 16 TOG INTERVAL	INTERV1	INTERV2	(Default = INTE
17Q IO SLOT 17 TOG INTERVAL	INTERV1	INTERV2	(Default = INT

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-9 — Instructions for MC Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
HG MC BOX POINT DISPLAY			
15 ***	TOG INTERVAL 1 (BOXTOG1)	Enter 0 to 9999 seconds 0 = Never shed	Time-Out-Gate (TOG) intervals 1 & 2 enable selection of either of two time-out intervals for each for each slot in this MC box. The time-out interval is the time between two successive transmissions from a supervisory controller to a particular slot in this box. If a time-out interval is exceeded, the respective slot is automatically placed in the manual mode.
16 ***	TOG INTERVAL 2 (BOXTOG2)	(Default = 0 seconds)	
(See Note)			
16A– 16Q	SLOT 1–16 TOG INTERVAL (TOGINTSL)	Circle one: INTERV1 INTERV2 (Default = INTERV1)	Specifies whether TOG INTERVAL 1 (INTERV1) or TOG INTERVAL 2 (INTERV2) entered on Lines 15 and 16 is to be the time-out gate interval for the particular slot.
17A– 17Q	IO slot 1-16 TOG INTERVAL	circle one: INTERV 1 INTERV 2 (Default = INTERV1)	Specifies whether TOG Interval 1 (INTERV1) or TOG Interval 2 (INTERV2) entered on lines 15 and 16 is to be the timeout gate interval for the particular IO slot.

AU, GPCI, H4500 & RCD BOX DATA POINT CONFIGURATION FORM**HG BOX DATA POINT DISPLA**

1	HIWAY NUMBE (HWYNUM)		(Default = 1)		
2	BOX NUMBER (BOXNUM)		(Default = 0)		
3	BOX TYPE (BOXTYPE)	CB	AU	HLPIU	LLPIU
		LEPIU	CBRCD	EC	ECRCD
		MC	MCRCD	DHP	OPSTA
		H4500	HG	GPCI	NOTCONFG
3A	BOX ASSIGNMENT (BOXASSN)	THISHG	REMOTEHG	ADDEDHG	4500

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

2.1.4.10 AU, GPCI, H4500, & RCD Box Data Point (Box/Slot) Configuration Form SW88-539

This form can be used to configure the box data point for an AU, GPCI, H4500, or RCD. It can also be used to delete a box data point. To delete a box data point, make entries in lines 1 and 2 as required, and then enter NOTCONFIG for line 3.

Table 2-10 contains instructions for Configuration Form *SW88-539*.

Table 2-10 — Instructions for Box Data Point Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HG BOX DATA POINT DISPLAY			
1	HIWAY NUMBER (HWYNUM)	Enter a number from 1 to 20 (Default = 1)	Hiway Number assignments to HGs are made in LCN node configuration.*
2	BOX NUMBER (BOXNUM)	Enter a number from 1 to 63. (Box number 4 is reserved for the Operator Station) (Default = 1)	Defines the box address of this box on the selected Data Hiway. If no Operator Station is on this hiway, box number 4 can be assigned to another box.
3	BOX TYPE (BOXTYPE)	Circle one: CB AU HLPUI LLPUI LEPIU CBRCD EC ECRCD MC MCRCD DHP OPSTA H4500 HG GPCI NOTCONFIG	Valid entries for this line are the following: AU = Analog Unit MCRCD = Multifunction Controller Reserve Controller Director CBRCD = Basic Controller Reserve Controller Director ECRCD = Extended Controller Reserve Controller Director

(Continued)

*Refer to Network Form SW88-509, Hiway Gateway Node Display

AU, GPCI, H4500 & RCD BOX DATA POINT CONFIGURATION FORM

HG BOX DATA POINT DISPLA

1	HIWAY NUMBE (HWYNUM)		(Default = 1)		
2	BOX NUMBER (BOXNUM)		(Default = 0)		
3	BOX TYPE (BOXTYPE)	CB	AU	HLPIU	LLPIU
		LEPIU	CBRCD	EC	ECRCD
		MC	MCRCD	DHP	OPSTA
		H4500	HG	GPCI	NOTCONFIG
3A	BOX ASSIGNMENT (BOXASSN)	THISHG	REMOTEHG	ADDEDHG	4500

References: Data Hiway Box/Slot, and Data Point Form Inst. in the Impl./Startup & Reconfig. binder.
Data Entity Builder in the Implementation/Engineering Operations-1 binder.

Table 2-10 — Instructions for Box Data Point Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
3A	BOX ASSIGNMENT (BOXASSN)	Circle one: THISHG REMOTEHG ADDEDHG H4500 (Default= THISHG)	OPSTA = Basic or Enhanced Operator Station H4500 = Honeywell 4500 Computer GPCI = General Purpose Computer Interface If CBRCD, ECRCD, or MCRCD is entered for this line, also fill out line 3A. Identifies which preferred access device is primarily responsible for operation of the type of box selected in line 3 (refer to Table 2-2 and to Section 2.1.3).

DATA POINT FORMS

Section 3

General instructions are provided in this section to fill out the Hiway Gateway Data-Point forms. You must also refer to the *HG Parameter Reference Dictionary* in the *Implementation/Hiway Gateway - 1* binder where detailed information about each Data-Point parameter is provided. The HG Data-Point Forms are located in the *Implementation/Configuration Forms* binder.

3.1 FORM TYPES

Forms are provided for each type of HG process data-point as follows:

<u>Box Type</u>	<u>Process Data-Point Type</u>
Basic Controller (CB)	Remote Variable (RV) Regulatory
Data Hiway Port (DHP) and IPCD 620 Hiway Interface Module (620 HIM)	Analog Composite Analog Input Analog Output Digital Composite Digital Input Digital Output Counter
Extended Controller (EC)	Digital Input Regulatory
Hiway Gateway (HG)	Batch Assign
Highway Gateway Library Forms are located in the <i>Implementation/Configuration Forms</i> binder.	
High-Level Process Interface Unit (HLPIU)	Analog Composite Analog Input Analog Output Digital Composite Digital Input Digital Output Counter (16-Bit) Counter (32-Bit)
Low-Energy Process Interface Unit (LEPIU)	Analog Input
Low-Level Process Interface Unit (LLPIU)	Analog Input

(Data-Point Forms, Continued)

Multifunction Controller (MC)

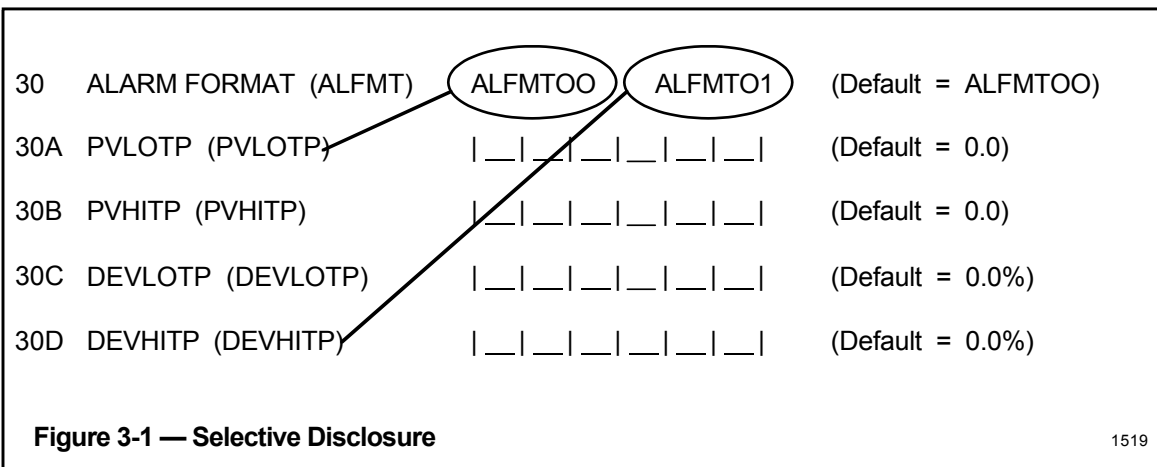
Analog Composite
 Analog Input
 Analog Input with Accum.
 Analog Output
 Digital Composite
 Digital Input
 Digital Output
 Counter (16-Bit)
 Process Module
 Regulatory
 Flag
 Numeric
 Timer
 Logic Block

3.2 GENERAL INSTRUCTIONS

3.2.1 Line Numbers

The Data-Point forms have also been designed so that the lines on them correspond with the Parameter Entry Displays (PEDs) at the Universal Station.

The line numbers may contain an alphabetic suffix letter as shown in Figure 3-1. An alphabetic suffix indicates that the line may be exposed or suppressed on the PED, depending on the selection made in a previous line.



In the example shown in Figure 3-1, selecting ALFMT00 in Line 30 causes Lines 30A and 30B to be disclosed on the PED, while Lines 30C and 30D are suppressed and are not shown. Conversely, selecting ALFMT01 in Line 30 causes Lines 30C and 30D to be exposed, while Lines 30A and 30B are suppressed. To determine which lines have to be filled in, refer to the definition of the ALFMT parameter in the *Hiway Gateway Parameter Reference Dictionary*, in the *Implementation/Hiway Gateway - 1* binder.

3.2.1.1 Regulatory Data Points

For regulatory data-point forms, an asterisk suffix for a line number indicates that the line may or may not have to be filled in by the user, depending on the algorithm selected for the regulatory data point. As shown in Figure 3-2, algorithm selection is accomplished at the ALGIDDAC parameter on the form, which provides the user with a choice of algorithms that are applicable to the respective highway box. For example, if Logic was selected as the algorithm for an Extended Controller (EC) Regulatory Data Point, the user would look up the definition of "Logic" in the *Hiway Gateway Parameter Reference Dictionary*, to determine which of the lines (parameters) have to be filled in on the form. In this example, the user would not fill in Lines 41 and 48 because the CTLACTN and PVTRACK parameters do not appear in the listing.

3.2.1.2 Extended Controller Regulatory Form

In addition, the EC Regulatory Form requires the user to write in the parameter name and value (or selection) for Lines 73 through 94, using the same technique as described in the previous paragraph except that the parameter names are taken from the parameter listing in the dictionary. The CB and MC Regulatory Forms do not require the user to write in the parameter names.

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EC REGULATORY DATA POINT CONFIGURATION FORM

HG POINT ASSIGNMENT DISPLAY

1 TAG NAME (Default = All Under-scores)

2 UNIT ID (UNIT) (Default = All Blanks)

3 POINT DESCRIPTOR (Default = All Blanks)

4 POINT KEYWORD (Default = All Blanks)

5 PRIMARY MODULE POINT ID (Default = All Under-scores)

6 HWAY NUMBER (HWAYNUM) (Default = 1)

7 BOX TYPE (PNTBOXTY) (Enter EC)

8 BOX NUMBER (BOXNUM) (Default = 00)

9 POINT DISCLOSURE (PTDISCAL) (Default = BRIEF)

10 SLOT NUMBER (SLOTNUM) (Default = 1)

11 LOAD DESTINATION (LOADDEST) (Default = HG)

12 ALGORITHM (ALGIDDC) (NOTE: For lines 41-106, refer to the name of the algorithm in the HG Parameters Manual to determine which lines have to be filled out.)

Form

ALGIDDC Parameter

Page of HG Parameter Reference Dictionary

LOGIC (EC)

Logic - Logic (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Logic** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data-entry displays. The outstanding characteristic of the **Logic** algorithm is that it provides logic computations that yield true/false responses.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
40 DISPLAY SUPPRESS LOW (SUPPLOW)	NOSUPP INSUPP OUTSUPP (Default = NOSUPP)	ALFMT (Alarm Trip-point Values)	SUPPIO CTLEQN (A,B)	NMODE
41* CONTROL ACTION (CTLEQN)	DIRECT REVERSE (Default = DIRECT)	ALPRIOR	OUTIND	NMODATTR
42* CONTROL EQUATION (CTLEQN)	BQA BCB BQC (Default = BQA)	CCRANK	PVSIGNAL	PSTMODE
43 OUTPUT INDICATION (OUTIND)	DIRECT REVERSE (Default = DIRECT)	CCPRIPNT	PVSLTSRC	MODEPERM
44 PV SIGNAL TYPE (PVSIGNAL)	OUTPUT PV HWAY LSP S1 S2 S3 S4 S5	EIPPCODE	SPSIGNAL	MODE
45 PV INPUT SOURCE (PVSLSRC)	NOTCONFG (Default = PV)	EIREVENT	SPSLTSRC	
46* SP SIGNAL TYPE (SPSIGNAL)	OUTPUT PV HWAY LSP S1 S2 S3 S4 S5	EIPENB	SRCHAR	
47* SP INPUT SOURCE (SPSLSRC)	NOTCONFG (Default = LSP)	ALMASK	SPFORMAT	
48* PV TRACKING (PTRACK)	NOTRACK TRACK (Default = NOTRACK)			

Lines with asterisks

Figure 3-2 — Determining Lines to be Filled in for a Regulatory Form

1520

3.2.2 Form Entries

There are two types of entries that you can make on the configuration forms as shown in Figure 3-3. Some parameters require that you write in a numeric value or alphanumeric character string in the spaces provided on the form. Other parameters require that you make a selection from the complete range of possible selections for the parameter, as if in a menu. To make a selection from this menu of choices, you can use a pencil or pen to circle the selection.

Written Entry

20 E.U. DESCRIPTOR |_|_|_|_|_|_|_|_| (Default=All Blanks)
(EUDESC)

21 CHARACTERIZATION LINEAR JTHERM KTHERM TTHERM
(PVCHAR) STHERM SQRROOT BURNSRTD RADIAMAT
(Default = LINEAR)

Circled Entry

Figure 3-3 — User Entries 1521

NOTE

Selecting Brief as the entry for the POINT DISCLOSURE (PTDISCL) parameter instead of Full, causes default values to be entered for those parameters not displayed by the DEB, and these values cannot be changed. Therefore, always select Full for the PTDISCL parameter for those data points where you decide to change the default value at a later time.

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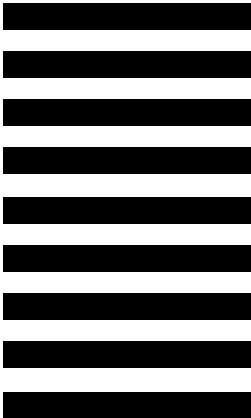
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